

END-OF-STUDY PROJECT REPORT

Submitted in Partial Fulfillment of the Requirements for the
BACHELOR DEGREE IN COMPUTER SCIENCE

Field of Study : Software Engineering and Information Systems

Hybrid Marketing Mix Modeling with Conversational AI for Insight Delivery

By
GHAZI MANAI AND EL MAHBOULI AZZA

Conducted within Technozor SARL



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Dedication

This thesis is dedicated to my beloved family, whose unconditional love and support have been my greatest strength.

To my mentors and my colleague, Azza Elmahbouli, for their invaluable guidance and collaboration.

To my friends, for their constant encouragement and uplifting presence throughout this journey.

This work is dedicated to all of you.

Ghazi MANAI

This thesis is dedicated to my beloved family, whose unwavering support and endless encouragement have been the cornerstone of my journey.

To my family, thank you for instilling in me the values of perseverance, patience, and ambition.

To my friends and loved ones, your belief in me gave me strength during moments of doubt.

And to my project partner, Ghazi MANAI, your dedication, teamwork, and friendship have made this experience both meaningful and memorable.

This accomplishment is as much yours as it is mine.

Azza EL MAHBOULI

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My heartfelt thanks also go to my colleague and project partner, **Azza ELMAHBOULI**. Her dedication, insight, and collaborative spirit have made this experience both productive and enjoyable. Sharing this journey with her has been a privilege.

To my **family**, thank you for being my constant source of strength. Your love, encouragement, and faith in me through every challenge and every success have given me the resilience to keep pushing forward.

Finally, to my **friends**, whose laughter, companionship, and moral support have brightened even the most stressful days I am deeply grateful. You remind me that success is sweeter when shared.

This accomplishment is not mine alone it belongs to all of you who walked this path with me. Thank you.

Ghazi MANAI

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General Introduction

In today's data-driven business landscape, making informed decisions requires more than just access to information; it requires tools that can extract actionable insights from complex datasets. Marketing Mix Modeling (MMM) has emerged as a key technique for quantifying the impact of marketing activities on sales and business outcomes. By statistically analyzing historical data, MMM enables businesses to evaluate the effectiveness of each marketing channel, optimize budget allocation, and forecast performance.

However, while MMM offers deep quantitative insight, its outputs are often presented in static reports, dashboards, or spreadsheets that are difficult for non-technical stakeholders to interpret. This disconnect can lead to underutilization of valuable analytical results or misinformed decisions.

To bridge this gap, intelligent systems — particularly AI-powered conversational agents — have become increasingly relevant. Intelligent chatbots now serve as interfaces that translate complex data into natural conversations, enabling users to ask questions and receive clear, contextual, and visual answers in real time.

In this report, we present the system we developed: a chatbot-based assistant designed to interpret and communicate the results of a Marketing Mix Modeling (MMM) model. While the core value lies in the MMM analysis itself, our assistant transforms its static outputs into an interactive dialogue, making insights more accessible and actionable for both technical analysts and business users.

At the heart of our project is a hybrid system that combines robust modeling techniques—such as Robyn MMM and linear regression—with advanced Natural Language Processing (NLP) and semantic search powered by vector databases. This integration enables users to interact with marketing data through natural language, making the interpretation of performance insights more accessible and actionable.

Finally, this project demonstrates how the integration of Marketing Mix Modeling with an AI-powered chatbot can transform complex analytical outputs into accessible, conversational insights. By developing a system that combines Robyn MMM, linear regression, NLP, and semantic search, we created a user-friendly assistant capable of interpreting and communicating marketing performance data in real time. Our solution bridges the gap between data science and business decision-making, making advanced analytics usable for a wider audience. Looking ahead, this approach opens new perspectives for expanding similar intelligent systems to other domains, enhancing data-driven decision-making across industries through natural, interactive interfaces.

The report is organized as follows:

- **Chapter 1: “Project Framework”** Describes the problem context, the motivations behind the project, and an overview of the host organization.
- **Chapter 2: “State of the Art”** Reviews existing technologies related to intelligent assistants and Marketing Mix Modeling, highlighting current gaps.
- **Chapter 3: “Data Collection and Preparation”** Details how the relevant structured and unstructured data were collected, cleaned, and transformed to support both the modeling and conversational interface.
- **Chapter 4: “Model Implementation”** Explains the architecture and construction of the hybrid MMM system and its integration with the chatbot to enable real-time, interactive understanding of campaign performance.

Chapter 1

Project Framework

1.1 Introduction

This chapter introduces the host organization and its current marketing analytics approach, then presents the proposed advanced solution to improve decision-making. It outlines the project’s main functional and non-functional requirements, followed by a summary of the two key components: Marketing Mix Modeling (MMM) and the intelligent chatbot development. Finally, it describes the chosen CRISP-ML methodology, which provides a structured framework to ensure the project’s success by aligning business goals with technical implementation.

1.2 Presentation of the Host Organization

Technozor SARL, founded in 2021, specializes in providing AI-driven solutions to address complex business challenges. With a team of data enthusiasts, engineers, and scientists, the company focuses on building enterprise AI systems that transform organizations through data analytics, machine learning (ML), and advanced technologies like anomaly detection, computer vision, and natural language processing (NLP). Technozor’s services cover various aspects of data strategy, management, and infrastructure, helping businesses design effective data strategies, automate operations, and optimize decision-making. They offer tailored solutions in areas like fraud detection, graph analytics, and mathematical optimization, empowering companies to predict outcomes, personalize services, and gain a competitive edge in the AI era.

1.3 Study of the Existing System

The company’s current analytical framework is based on early-generation MMM techniques¹ rooted in the foundational works of Little (1970) and Kotler (1971). These classical approaches typically employ linear regression models and heuristic evaluations for marketing decision-making.

¹Chapter two will examine the development and limitations of early-generation MMM approaches.

1.4 Proposed Solution

To address these issues of such system, we propose an hybrid Marketing Mix Modeling solution combining advanced and traditional methods, approved by an analytical chatbot.

1.5 Functional and Non-Functional Requirements

This section outlines the functional and non-functional requirements that define the expected capabilities, performance, and quality standards of the system throughout its development and deployment.

1.5.1 Functional Requirements

The following specifications outline the essential features that the system must have in order to be deemed comprehensive and efficient:

- **F1 – Data ingestion and preprocessing:** Uploading and cleaning marketing datasets (managing missing values, formatting dates, and normalizing variables) must be supported by the system.
- **F2 – Feature engineering:** The system must allow automatic generation of features such as lagged variables, weighted variables, and categorical encodings.
- **F3 – Model training:** The system must be able to train both Robyn MMM and linear regression models using historical marketing data.
- **F4 – Model evaluation:** The system must compute and display performance metrics such as RMSE, R^2 , and MAE for both models.
- **F5 – Budget simulation:** The system must allow users to simulate marketing budget allocation scenarios and observe predicted outcomes.
- **F6 – Variable importance analysis:** The system must extract and display the contribution of each marketing channel to overall performance.
- **F7 – Chatbot interaction:** Users must be able to ask the chatbot questions in natural language and receive relevant insights based on the model results.
- **F8 – Deployment interface:** A user-friendly web interface must allow interaction with the model and chatbot outputs.

1.5.2 Non-Functional Requirements

These specifications guarantee the system's effective operation as well as its dependability, security, and maintainability:

- **NF1 – Usability:** The interface must be intuitive and accessible to both technical and non-technical users.
- **NF2 – Performance:** The system must respond to user queries (including chatbot interactions) within a few seconds.

- **NF3 – Scalability:** The system architecture must allow for future integration of more advanced models or larger datasets.
- **NF4 – Maintainability:** The codebase must follow modular programming principles to facilitate future updates and debugging.
- **NF5 – Interpretability:** All outputs, especially those from Robyn MMM, must be accompanied by clear explanations, either via the UI or chatbot.
- **NF6 – Security:** The system must securely handle sensitive business data, with proper access control and data privacy measures.
- **NF7 – Reliability:** The system must ensure consistent performance across different datasets and user sessions, with minimal errors or crashes.

These requirements provide a clear foundation for the system’s development, guiding both technical implementation and user experience design in the later phases of the project.

1.6 Project Pipeline

To better understand how these requirements are realized in practice, the project pipeline is divided into two primary parts: **Marketing Mix Modeling (MMM)** and **Intelligent Chatbot Development**. The main stages and methods employed during the project are described in the following subsections.

1.6.1 Marketing Mix Modeling (MMM)

Creating a reliable and intelligible model for evaluating the effectiveness of marketing campaigns is the aim of this project phase. It goes over each step, from cleaning raw marketing data to using complex modeling techniques and comparing the results.

Data Collection and Preparation

- Explanation of marketing data variables (SEM, SNS, CV, etc.)
- Data cleaning: missing value handling, type conversion, date formatting
- Normalization using min-max scaling
- Feature engineering:
 - Lag feature generation
 - Weighted feature creation using regression coefficients
 - Dummy variable encoding and grouping campaigns

Model Development

In order to measure the impact of marketing initiatives and evaluate their efficacy, this phase focuses on applying two complementing modeling approaches: Robyn MMM and conventional linear regression.

Model Comparison and Hybrid Approach

- Comparative study of both models on key metrics
- Decomposition and channel contribution analysis
- Hybrid use of both models to ensure robustness and interpretability

1.6.2 Intelligent Chatbot Development

This project's main goal is to provide an approachable conversational assistant that facilitates the interpretation and investigation of the MMM models' output. Users, particularly non-technical stakeholders, can ask questions, visualize insights, and make well-informed decisions thanks to the chatbot's integration of natural language comprehension with both organized and unstructured data.

Data Preparation for Chatbot

The following steps summarize the process we followed to prepare both structured and unstructured data for integration into the chatbot's semantic search and question-answering capabilities.

- Structured data: 7 CSV files with model outputs and KPIs
- Unstructured data: 5 PDFs with marketing concepts and technical guides
- Text preprocessing: cleaning, chunking, and metadata tagging
- Embedding generation using Google Generative AI
- Storage in ChromaDB for semantic search

Architecture and Functionality

The following points highlight the main components that power the chatbot's ability to understand queries and deliver accurate responses.

- Dual-retrieval architecture (semantic + SQL queries)
- Integration of vector search and structured reasoning
- LangChain for hybrid tool orchestration

User Interaction and Use Cases

The following use cases illustrate how users interact with the chatbot to explore model results, understand key metrics, and simulate marketing scenarios through a conversational interface.

- Natural language queries and intelligent responses
- Explanation of technical terms (e.g., NRMSE)
- Visual chart generation (e.g., ROI, contribution)
- Budget simulation interpretation

Deployment and Testing

The points below summarize how the system was deployed, tested with users, and iteratively improved based on feedback to ensure reliability and usability.

- Local deployment using Streamlit interface
- User testing and feedback loop
- Continuous updates and improvement cycles

1.7 Used methodology

Since a methodology offers a clear structure and framework, it is essential to the success of our project.

It provides a number of benefits, such effectively managing our data and time by outlining precise procedures, preserving consistency, and producing work of a high caliber.

A disciplined methodology is vitally necessary for the success of our AI-based project in this environment. The Cross Industry Standard Process for Machine Learning (CRISP-ML) methodology was used for this reason.

Because of its emphasis on business insight, iterative nature, and compatibility with data science projects, it guarantees that, during the course of the project, our AI solution successfully meets actual business needs.

As seen in this Figure1.1, there are six steps in this procedure.

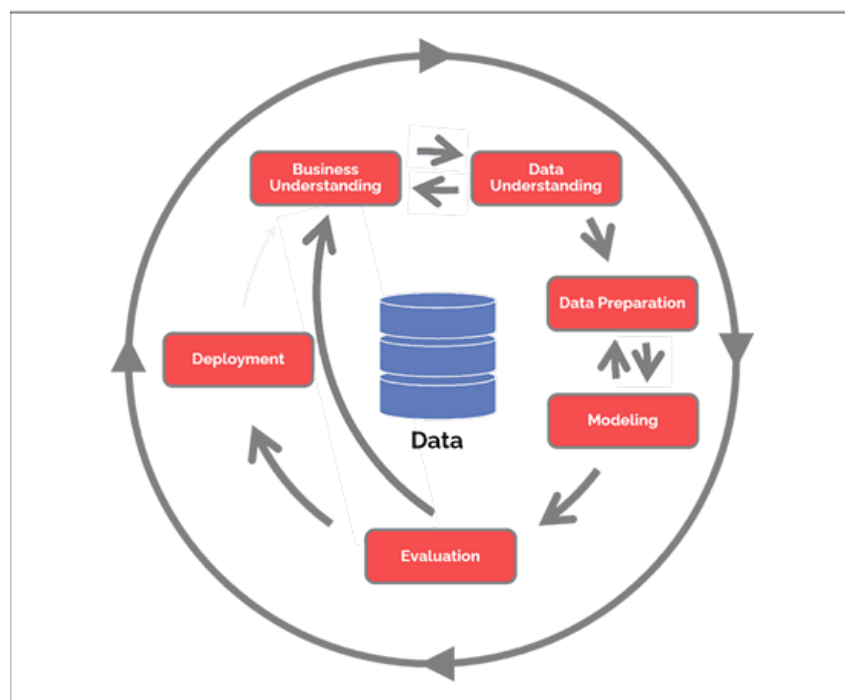


Figure 1.1: Crisp-ML life cycle [2]

- Business Understanding is the initial stage, during which the issue that needs to be resolved for our company is identified.
- Data Understanding, the second phase, is broken down into two phases. To find out what data is available and how to obtain it, data is first searched. Second, the data is examined and its condition assessed.
- The process of organizing data to meet modeling requirements is known as data preparation.
- Modeling: During this stage, the various modeling options are investigated.
- To make sure that project goals are met prior to deployment, the predictive model's performance is assessed during the evaluation phase using a set of criteria.
- Lastly, the model is made available to users during the deployment phase. There are several ways to accomplish this, such as using an API or a graphical user interface.

1.8 Conclusion

This chapter has laid the foundation of the project by presenting the host organization's current challenges and the motivation for adopting advanced marketing analytics. We have defined the project's scope, goals, and key requirements, emphasizing the dual focus on Marketing Mix Modeling and an intelligent chatbot to improve data interpretation for non-technical users. The chosen CRISP-ML framework ensures a systematic approach to development, bridging business objectives and technical execution.

The following chapter will provide a detailed state-of-the-art review, exploring the evolution of Marketing Mix Modeling techniques, their limitations, and modern advancements such as the Robyn framework. It will also examine the integration of AI technologies—including NLP, vector databases, and chatbots—that underpin the proposed solution.

Chapter 2

State of the Art

2.1 Introduction

In this chapter, we examine the technological evolution of Marketing Mix Modeling (MMM), starting from its classical linear regression foundations to the emergence of advanced, automated solutions such as Meta’s Robyn. We analyze the analytical, operational, and strategic limitations of traditional MMM and how modern frameworks address these challenges through techniques like adstock transformations, saturation effects, and hyperparameter optimization. Furthermore, the chapter explores the integration of artificial intelligence—particularly natural language processing (NLP), semantic search, and vector databases—to enable more dynamic and scalable interpretations of model outputs. These advancements lay the groundwork for developing intelligent systems such as chatbots capable of interacting with MMM results in a user-friendly, explainable, and data-driven manner.

2.2 Early Generation of MMM

Marketing Mix Modeling (MMM) is an analytical method used to measure how different marketing efforts and external influences affect sales and other key business results. This approach helps companies better understand the effectiveness of their marketing investments and make more informed decisions to optimize their overall strategy.

In early MMM implementations, linear regression was the dominant modeling approach due to its simplicity, interpretability, and alignment with classical econometric practices.

Marketing data like spend, conversions, and campaigns were stored in spreadsheets, but collecting and combining this data was manual and time-consuming.

This chapter presents an overview of both the traditional and modern MMM methods, outlining the limitations of the early approach, and introducing a new approach that focuses on Robyn as a robust alternative in combination with linear regression :

2.2.1 Traditional Linear Regression Framework

The foundational structure of early MMM is the multiple linear regression (MLR) model, which aims to establish a relationship between marketing inputs and a target Key Performance Indicator (KPI), such as revenue or conversions:

$$Y_t = \beta_0 + \sum_{i=1}^n \beta_i X_{i,t} + \varepsilon_t \quad (2.1)$$

Where:

- Y_t : dependent variable (conversion) at time t
- $X_{i,t}$: independent marketing variable i at time t
- β_i : regression coefficients
- ε_t : error term

Our dependent variable conversion represented by Y can be an action of a purchase, download, or sign-up made by the consumer.

The independent variables represented by X in this model are spends on TV ads, radio ads, digital ads, price promotions, and seasonality indices.

Among the characteristics of this model, the following can be distinguished:

Data Splitting: Training and Testing Sets :

To evaluate model performance and generalizability, the data is split into:

- **Training set (70%)** — used to fit the regression model.
- **Testing set (30%)** — reserved for validating model predictions on unseen data.

This common practice helps prevent overfitting and provides an unbiased estimate of how well the model might perform on future data.

Model Training and Prediction :

The linear regression model is trained using the training data, estimating coefficients that quantify the marginal impact of each independent variable on the dependent variable. The fitted model then predicts outcomes on the test data, allowing direct assessment of predictive accuracy.

Model Evaluation Metrics

To quantify the model's predictive performance, three key metrics¹ are calculated on the test set:

- **Root Mean Squared Error (RMSE):** Measures average prediction error magnitude, with higher sensitivity to large errors.

¹Each metric's equation is detailed in Chapter 3, Section 3.4 "Key Performance Metrics."

- **Mean Absolute Error (MAE):** Measures average absolute deviation between predicted and actual values, providing an intuitive error scale.
- **Coefficient of Determination (R^2):** Indicates the proportion of variance in the dependent variable explained by the model, reflecting overall fit.

These metrics together provide a comprehensive view of both error magnitude and explanatory power.

Variable Importance Extraction :

The model extracts the regression coefficients (β values) for each predictor. By taking the absolute value of these coefficients, the function creates a ranking of variables by their estimated impact magnitude on the dependent variable.

In MMM, understanding variable importance is critical for budget allocation and strategic decision-making, as it highlights which marketing channels or contextual factors most strongly influence outcomes.

Output Summary :

Finally, the function returns a comprehensive set of results including:

- The fitted linear regression model object.
- Predicted values on the test set alongside actual observed values.
- Performance metrics (RMSE, MAE, R^2).
- Variable importance ranked by coefficient magnitude.
- Full statistical summary of the model (e.g., coefficient estimates, standard errors, significance levels).

This detailed output supports both quantitative assessment and qualitative interpretation, which is vital for effective marketing strategy optimization.

2.2.2 Limitations of Traditional MMM

Despite being foundational, early MMM models have inherent limitations:

Analytical Limitations

The analytical limitations in this MMM are weaknesses in its assumptions, data, or structure that reduces its accuracy, generalizability, or interpretability as shown below :

- **No modeling of lagged effects:**

The current approach does not account for lagged effects such as adstock (Broadbent, 1997) , which are crucial for capturing the prolonged impact of marketing actions.

- **The assumption of a linear relationship:**

By assuming that for each increase in marketing spend there is a proportional increase in return on investment (ROI), the model fails to capture saturation or diminishing returns.

- **No cross channel interaction modeling:**

The model doesn't capture how different marketing channels, like TV ads or digital ads, influence each other. This makes it harder to understand how these channels work together or interfere with each other, which is especially important in today's complex digital world where many channels are used at once.

Operational Limitations

we are gonna discuss below the practical constraints that affect the implementation, scalability, or usability of the model in real-world business settings :

- **Manual data processing:**

Data preparation relies on manual steps, slowing analysis and increasing the risk of errors, while modern practices emphasize automation and scalability.

- **Subjective interpretation:**

The absence of a rigorous analytical framework allows for decisions influenced by cognitive biases, reducing objectivity and reproducibility.

- **Low responsiveness:**

The current system doesn't take advantage of the quick adjustments possible in digital marketing. This limits the ability to make real-time changes to campaigns based on their performance.

Strategic Limitations

In here we refer to the constraints that reduces the model's ability to support long-term, high-level marketing decisions aligned with broader business goals.

- **Short-term optimization focus:** The current model emphasizes immediate results, neglecting long-term impacts on the brand or customer loyalty due to the lack of predictive modeling.

- **No scenario simulation:**

unlike the modern approaches that allow companies to test the impact of different budget allocations, the existing system doesn't support predicting results when the marketing budget is spent differently.

2.3 Robyn: A Modern Approach to MMM

Robyn is an open-source R package developed by Meta (formerly Facebook) and initially released in 2021. It was designed as a comprehensive, automated solution dedicated entirely to Marketing Mix Modeling (MMM).

It aims to reduce human bias by means of machine learning techniques like ridge regression and evolutionary algorithms for hyperparameter optimization . It enables actionable decision making by providing a budget allocator and diminishing returns curves, and it supports ground-truth calibration to account for causation.

Here are a detailed explanation of Robyn MMM key components As described in the Robyn Analyst's Guide [4] and feature documentation [5],

2.3.1 General Robyn MMM Equation

The base model in MMM is typically specified as a regression model:

$$Y_t = \beta_0 + \sum_{i=1}^k f_i(X_{it}) + \sum_{j=1}^m \gamma_j Z_{jt} + \varepsilon_t \quad (2.2)$$

Where:

- Y_t : Dependent variable at time t (conversions Y)
- β_0 : Intercept, captures the baseline level of conversions Y_t when all media X_{it} and control variables Z_{jt} are zero.
- X_{it} : Spend for the i^{th} paid media channel at time t
- $f_i(\cdot)$: Non-linear Transformation function for X_{it} (adstock + saturation)
- Z_{jt} : Control variables wick are organic Media variables, Contextual variables, and seasonal variables.
- ε_t : Error term

Each of these Model components is gonna be detailed below .

2.3.2 Variable Categorization

In terms of what data sets to collect, it's important to distinguish between dependent and independent variables for the Marketing Mix Model (MMM).

Dependent Variable Y

The dependent variable is the primary KPI or metric that the MMM aims to explain or predict. The dependent variable we used for static Robyn MMM is:

- **Conversion** : a conversion is a measurable action taken by a consumer that aligns with a business objective, such as a purchase, download, or sign-up.

Independent Variables

Independent variables are factors that are assumed to influence the dependent variable. These can be grouped into the following categories:

- **Paid Media X:** Includes media channels with associated marketing spend, such as TV, online ads, etc. These variables are typically subject to transformation techniques like adstock (to model carryover effects) and saturation (to model diminishing returns).
- **Organic Media Z:** Includes marketing efforts without direct spend, such as newsletters, push notifications, and social media posts.
- **Contextual Variables Z:** External factors that may affect the dependent variable, such as macroeconomic indicators or competitor activity. These variables are included using the `context_vars` parameter.
- **Seasonal Variables Z:** Capture recurring patterns and trends like holidays, seasonal events, and weekday/weekend effects. These are often modeled using tools like Facebook’s Prophet.

This classification ensures that each variable type is treated with the appropriate modeling technique, leading to more accurate and interpretable results.

2.3.3 Non-linear Transformation

In this section we are gonna detail each of the non linear functions in robyn wich are adstock and saturation.

Adstock Transformation

Adstock is a mathematical transformation applied to media variables to reflect the carry-over effect of advertising — If you see an ad for a new phone today, you might not buy it immediately. But you remember it, think about it for a few days, and maybe buy it next week. That delay where the ad is still influencing your decision after it ran is the carryover effect. this is what adstock models.

Geometric Adstock

Geometric adstock is a method used in Marketing Mix Modeling to capture the carryover effect of advertising.

It assumes that the impact of media exposure does not disappear immediately but decays gradually over time.

This effect is modeled using a recursive formula where the current period’s adstock value is the sum of the current media input and a proportion of the previous period’s adstock.

Mathematically, it is expressed as:

$$\text{Adstock}_t = x_t + \theta \cdot \text{Adstock}_{t-1}$$

Where:

- x_t is the media spend or impressions at time t ,

- $\theta \in [0, 1]$ is the **decay rate** (adstock parameter),
- Adstock_t is the transformed media value at time t .

As seen in this Figure2.1

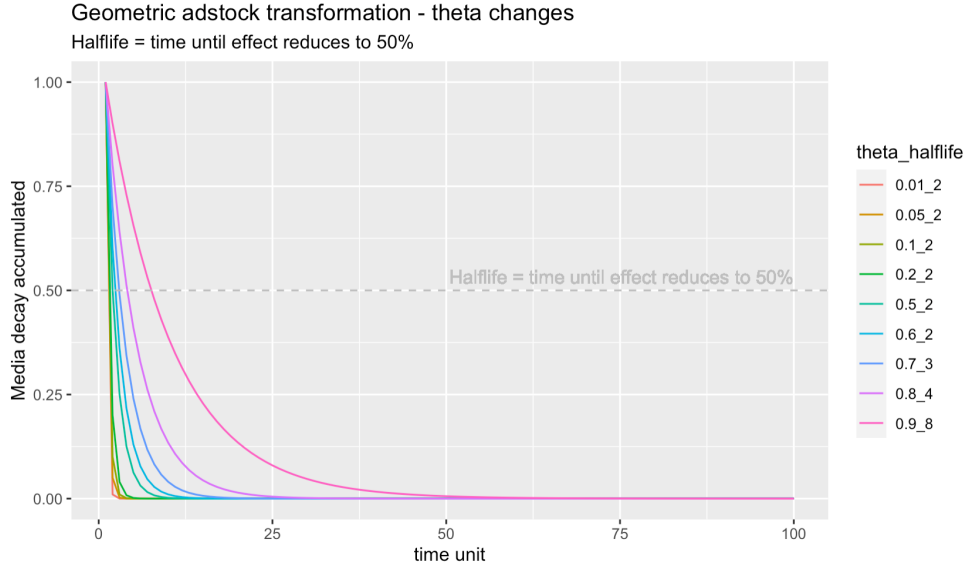


Figure 2.1: Geometric Adstock [2]

Saturation Modeling

Saturation in Robyn models the idea that increasing advertising spend does not lead to a proportional increase in conversions or sales.

At a certain point, additional spend has diminishing returns — meaning it becomes less effective.

Robyn applies a non-linear transformation (using a Hill function) to capture this behavior and reflect the real-world limits of media effectiveness :

$$\text{media_saturated}_j = \frac{1}{1 + \left(\frac{\gamma_j}{\text{media_adstocked}_j} \right)^{\alpha_j}}$$

where:

- media_saturated_j : The saturated response of the media channel j , capturing the diminishing returns effect after accounting for adstock.
- media_adstocked_j : The adstocked (carryover-adjusted) spend or exposure for media j , representing the effective media input after considering lagged effects.
- γ_j : The scale parameter controlling the horizontal position of the saturation curve; it determines the spend level at which saturation effects become significant.
- α_j : The shape parameter controlling the steepness of the saturation curve; it governs how quickly diminishing returns set in as media spend increases.

As seen in this Figure2.2,

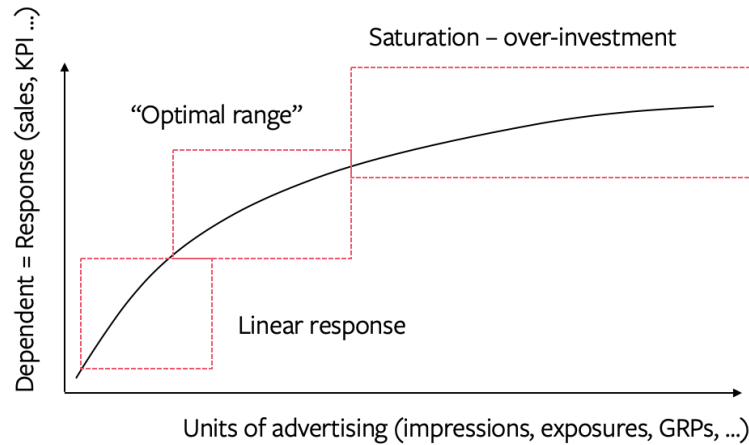


Figure 2.2: Saturation [2]

2.3.4 Multi-Objective Hyperparameter Optimization with Nevergrad

When facing multiple model candidates, it's common to prefer a model that's more plausible, one with slightly less fit but better alignment with current spend allocation. In other words, a good MMM needs to be both predictive and interpretable.

The implementation of multi-objective hyperparameter optimization is considered the most important innovation in Robyn.

Robyn uses Nevergrad, Meta's gradient-free optimization platform to perform this task with its so-called "ask & tell" interface. Simply explained, Robyn "asks" Nevergrad for the mutating hyperparameter values by "telling" it which values have better scores (objective functions).

Objective Functions

Each metric's equation is detailed in Chapter 3, Section 3.4 "Key Performance Metrics."

Robyn implements three objective functions² as the goals for hyperparameter optimisation:

- **NRMSE** Normalized Root Mean Square Error
- **DECOMP.RSSD**: Decomposition Root Sum of Squared Distance
- **MAPE.LIFT**: Mean Absolute Percentage Error

Hyperparameters

There are four types of hyperparameters in Robyn at the moment:

- **Adstocking**: θ when selecting Geometric adstocking.
- **Saturation**: α and γ parameters for the Hill function.

²Each metric's equation is detailed in Chapter 3, Section 3.4 "Key Performance Metrics."

- **Regularization:** λ for the penalty term in ridge regression.
- **Time series validation:** `train_size` for the percentage of training data.

2.3.5 Model Clustering

Robyn uses bootstrapping to calculate the uncertainty of CPA (Cost Per Acquisition). To further reduce the number of candidate models, k -means clustering is applied to the set of Pareto-optimal models.

- k -means clustering is performed on all Pareto-optimal model candidates to identify clusters of models with similar hyperparameter configurations. The parameter k represents the number of clusters.

2.3.6 Model Onepager

Robyn automatically exports a variety of charts and tables to a chosen folder to help understand the results and do further analysis. After running `robyn_outputs()`, it also creates a one-page summary for each best-performing model as shown in the annex 4.7. Below are explanations of key charts and outputs :

Chart Type	Description	Purpose/Insights
Response Decomposition Waterfall	Breaks down how much each variable (marketing channels, external factors) contributes to the total outcome (e.g., conversion)	Visualizes the relative contribution of each factor to the final result
Actual vs. Predicted Response	Compares actual outcomes (e.g., real sales) with what the model predicted	Evaluates model fit quality (e.g., R-squared near 0.9 indicates a very good fit); helps identify unusual events outside the model's data (like unexpected sales spikes)
Share of Spend vs. Share of Effect	Compares money spent on each paid media channel versus impact they had on results	Shows return on investment (ROAS) for each channel; identifies which channels provided better value; focuses only on paid media variables
In-Cluster Bootstrapped Confidence Interval	Uses bootstrapping to estimate confidence levels for metrics like ROAS or CPA	Provides 95% confidence intervals for each cluster of similar models; helps understand the range of expected performance
Adstock Decay Rate	Shows how long the effect of each media channel lasts over time (carryover effect)	Illustrates the decay rate for each channel; higher rates mean longer-lasting impact (offline channels like TV typically have stronger carryover)
Immediate vs. Carryover	Separates media impact into immediate effect (current period) and carryover effect (influence from past spend)	Distinguishes between direct impact of current spending and lingering effects from previous periods
Saturation Curves	Shows how media effectiveness diminishes as spending increases (diminishing returns)	Helps optimize budget allocation by identifying overspending (flat part of curve) or opportunity for more spend; grey area represents carryover effect, dot shows total effect
Fitted vs. Residual	Plots predicted values against leftover errors	Validates model quality; good models show randomly scattered points around zero with no clear pattern; patterns suggest missing variables

Table 2.1: Robyn Analytics Charts Reference

2.3.7 Budget Allocation

Robyn's budget allocator helps you decide how to split your marketing budget across channels in two ways:

- **Maximize Response:** If you know your total budget and have some rules for how much you can spend on each channel, Robyn figures out the best way to divide that money so you get the biggest overall impact. For example, “If I have \$X to spend, how should I split it across channels to get the most results?”
- **Hit a Target Efficiency (like ROAS or CPA):** If you want to reach a specific efficiency goal—say, a certain return on ad spend (ROAS) or cost per acquisition (CPA)—Robyn calculates both how to divide your budget and how much total budget you need to meet that goal. For example, “How much should I spend, and where, to hit a ROAS of Y?”

2.4 AI and Chatbots

Chatbots driven by AI mark a substantial advancement in human-computer interaction. To engage people in meaningful interactions, they make use of developments in reasoning and Natural Language Processing (NLP).

These systems can carry out activities, provide content summaries, respond to inquiries, and assist in decision-making. Large Language Models (LLMs) and AI agents enable modern chatbots to grasp intent dynamically, get pertinent facts, and provide responses that resemble those of a human, going beyond programmed responses.

They are effective tools in corporate settings because they can operate with both internal and external data sources, such as sales records, PDFs, websites, and knowledge bases. The chatbot is essential to our project since it helps non-technical people understand the results of Marketing Mix Modeling. It makes insights understandable and useful by converting static analytical results into interactive discussions.

2.5 Natural Language Processing (NLP)

The goal of the computer science and artificial intelligence discipline known as natural language processing (NLP) is to enable computers to comprehend and utilize human language. Enabling machines to read, write, talk, and comprehend language like people is the aim.

NLP is used in our project to help the assistant comprehend user inquiries and locate pertinent information in databases or documents. Recent advancements have made it possible for machines to comprehend language with far greater context and significance.

2.6 Vector Databases and Semantic Search

Direct keyword matching is usually the foundation of traditional search systems. Although this approach is simple, it frequently falls short when people use various words or phrase to convey the same idea. Our project incorporates a semantic search strategy, which prioritizes the text’s meaning over its word choice, to solve this problem. Vector representations of text, in which every bit of data is represented as a high-dimensional numerical vector, are the foundation of semantic search. Similar concepts can be recognized even if they are expressed differently because to these vectors, which capture the contextual meaning of the information.

In our implementation, we adopted the following tools:

- **ChromaDB** acts as the vector database for these embeddings, managing and storing them. We chose ChromaDB because of its adaptable and lightweight architecture, which makes it perfect for serverless, local deployments. It provides good performance for medium-sized applications without requiring cloud infrastructure, is open-source, and is simple to set up.
- **Google Generative AI Embeddings** were employed in order to convert textual data into vector form. Google’s sophisticated language models, which provide superior semantic understanding, generate these embeddings. An important benefit for our project was their ease of integration through API and support for multilingual input. Furthermore, the smooth integration of vector storage and embedding generation was made possible by their compatibility with ChromaDB.

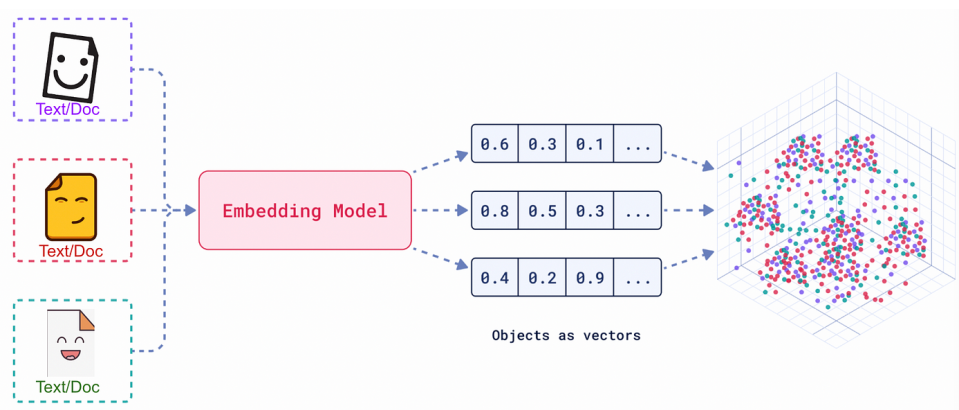


Figure 2.3: Embedding Model [3]

This combination greatly enhanced the chatbot’s comprehension of user inquiries by allowing us to create a semantic search engine that returns pertinent results based on conceptual similarities rather than exact word matches. This architecture demonstrates how our chatbot system incorporates semantic search to improve comprehension and response precision. Even if the user’s terminology deviates from the original material, the system can still obtain and utilize pertinent information by integrating ChromaDB with a language model.

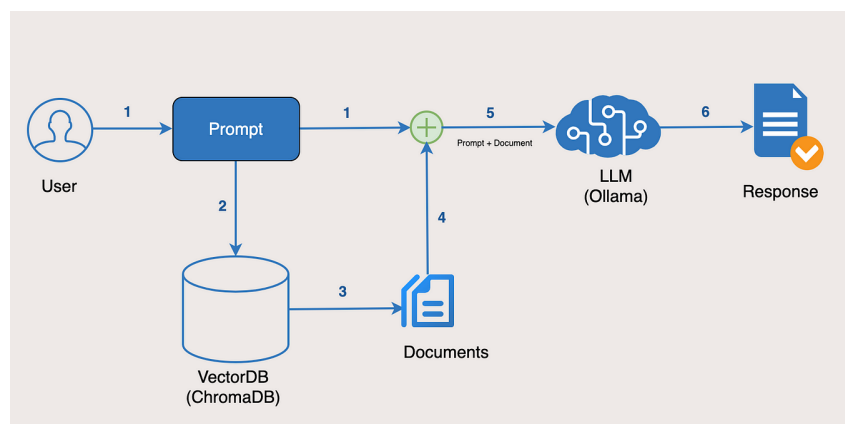


Figure 2.4: Semantic search and response generation using ChromaDB and a language model. [1]

2.6.1 Hybrid Reasoning: Combining Tools for Better Results

The assistant use a technique known as **hybrid reasoning** to give concise and beneficial responses. This implies that, depending on the kind of query the user poses, it integrates language comprehension with various tools, like databases and document search. This enables the assistant to respond with text, charts, or definitions based on the situation.

Use of LangChain

We used a framework called **LangChain** to enable this. LangChain facilitates the integration of language models with resources such as databases, visualization libraries, and PDF search engines. It serves as a clever link between actions and natural language. In response to a straightforward user query, the assistant can now generate visual charts, look up answers in files, and define important terms thanks to LangChain. Additionally, LangChain facilitates reasoning by automatically selecting the appropriate tool. For instance, the database will be accessed if the query concerns sales figures. It will look inside PDF documents if it's concerning a marketing notion. The fundamental concept of hybrid reasoning is this adaptability.

In Summary , The assistant can provide helpful responses in a variety of ways by fusing language comprehension, database access, and document search. The technique is useful for marketing and sales analysis because of this. Asking inquiries and receiving intelligent, precise answers is possible even for non-technical people.

2.7 Choosing the Right Language Model

2.7.1 Evaluation and Final Choice

SentenceTransformers, OpenAI (GPT-3.5/4), and Google's Gemini 2.0 are the three primary technologies we examined in order to develop our intelligent assistant. When it comes to language comprehension, usability, and affordability, each one has unique advantages.

- **SentenceTransformers** is an offline library suitable for simple tasks. It uses models such as BERT and RoBERTa to generate sentence embeddings. It doesn't require an internet connection and is free, but it doesn't produce any responses and has a limited ability to comprehend context.
- **OpenAI's GPT-3.5/4** is an extremely potent linguistic model. It comprehends difficult queries and provides superb responses. It is more difficult to modify or utilize locally, as it is only accessible online via a paid API. Additional configuration is needed for integration with other programs, such as ChromaDB.
- **Gemini 2.0 (Google Generative AI)** provides extremely high-quality language generation and understanding. It readily interacts with ChromaDB, supports a wide variety of languages, and retains context. Compared to OpenAI, it requires less manual setup and is free to use with a Google account.

Based on the primary factors we assessed, the following table lists the main distinctions between these three technologies:

Feature	Sentence Transformers	OpenAI GPT-3.5/4	Gemini 2.0 (Google)
Embedding Quality	Medium	High	Very High
Offline Usability	Yes	No	No
API Access	Not needed	Required	Required (easy)
Multilingual Support	Limited	Medium	Excellent
Cost	Free	Paid	Free
ChromaDB Integration	Compatible	Needs work	Seamless
LLM Response Quality	N/A	Excellent	Excellent
Context Retention	Limited	High	High

Table 2.2: Comparison of three language technologies tested

We selected **Gemini 2.0** for our project after testing and contrasting the three. The following factors led to this decision:

- When it came to comprehending questions and producing responses, it produced the highest-quality outcomes.
- With ChromaDB, which we utilized to store and search documents, it operated without a hitch.
- Compared to the other tools, it was better at handling various languages and technical jargon.
- In contrast to OpenAI, which needed a purchased key, it was simple to use and free with a Google account.
- Compared to SentenceTransformers, it offered more robust embeddings, which enhanced the assistant’s comprehension of intricate queries.

This enabled us to create a quick, intelligent, and reasonably priced assistant that could accurately interpret data and respond to customer inquiries.

2.8 Conclusion

This chapter has outlined the theoretical and technological foundations of modern Marketing Mix Modeling, with a particular focus on the Robyn framework and its advanced modeling techniques. It also explored how recent advancements in AI, NLP, and vector databases enhance interpretability and automation through intelligent systems like chatbots. These insights form the groundwork for the practical implementation phase. In the next chapter, we move from theory to practice by detailing the processes of data acquisition, cleaning, normalization, and feature engineering essential for building a robust MMM pipeline and chatbot assistant.

Chapter 3

Data Collection and Preparation

3.1 Introduction

This chapter presents the essential steps taken to construct a clean, structured, and analysis-ready dataset for Marketing Mix Modeling and chatbot integration. It begins by detailing the collection of marketing-related data, followed by rigorous cleaning to address inconsistencies and missing values. Subsequently, data preparation techniques such as min-max normalization and feature engineering are applied to enhance model performance. After preparing the data for Marketing Mix Modeling, the key performance metrics used to evaluate the model are discussed in detail, including the equations for each metric. These metrics not only reflect the effectiveness of the MMM but also serve as important input data for the chatbot. Finally, a specialized data preprocessing pipeline is implemented to support the chatbot's ability to interpret and respond accurately based on the dataset. This foundational stage ensures the reliability and consistency of subsequent modeling and interaction phases..

3.2 Data Collection

3.2.1 Marketing data

The data used in this project for both robyn and linear regression model was shared by a Japanese company on the internet for educational purposes. The dataset covers expenditures across different types of channels and marketing activities, collected daily over the year from March 8, 2023, to December 2, 2023. The marketing data is explained in detail below:

- **CV**: Represents conversions.
- **SEM (Search Engine Marketing)**: Search engine advertising spend.
- **ADNW1, ADNW2, ADNW3**: Spend on various digital ad networks.
- **SNS (Social Network Services)**: Spend on social media platforms.
- **AFF (Affiliation)**: Affiliate marketing spend.
- **DOUGA**: Investments in video platforms.

- **BRAND:** Brand awareness activities.
- **GT:** Company-specific contextual variable.

This period captures key seasonal events and marketing activities that allows us to analyze their impact on business performance throughout the year. It includes:

- **GAMEPR230324, GAMEPR230727:** Game-related public relations efforts.
- **INFLU230402, INFLU230404, INFLU230408, INFLU230821:** Influencer campaigns.
- **CP230317, CP230421, CP230811:** Specific promotional campaigns.
- **obon230812:** Obon holiday in Japan.
- **GW230429, GW230430:** Golden Week in Japan.

3.2.2 Data Cleaning

After assuring the loading of the dataset according to the three format we choosed csv, xslm and rds, we then pass to the most critical step in ensuring the integrity and consistency of our data , the cleaning process. In this process we performed several key operation such as adresssing missing values,convert data types, and ensuring proper structure as explained bellow :

NA Removal and Handling:

in this section we begin by ensuring that our dataset conatins our most necessary coloumn, the DATE ,wich is crucial for time series analysis later on.

If this column is missing, the function stops and throws an error to prevent further analysis. then during our next step data conversion there's a possibility of creation of NA values to prevent the function issues a warning to call for further investigation of the data.

Numeric Coercion:

in this step the script process all non-date columns and convert them to numeric values ensuring that non-numeric characters like commas and spaces are dealt with .

Date Formatting:

this is another step to ensure that the date columns is of the correct type (Date), if not we use the as.Date() function.

3.3 Data Preparation

3.3.1 Normalization using min-max scaling

A min-max normalization was applied to numeric predictors using the preProcess() function from the caret package. This transforms each feature to a $[0,1]$ range . the normalized data is then used in a linear models that aims to extract the contributing weight of each marketing channel to conversions by scaling the coeifficients to sum to 1 for later future engineering.

3.3.2 Feature engineering

In this project, various feature engineering techniques were applied to enhance the predictive power of the MMM models, aiming to regroupe marketing activities. Below are the details of each technique along with their corresponding mathematical formulations:

Lag Feature Generation

A lag feature represents the delayed impact of a variable over time. In this project, lag features for SEM (search engine marketing) were created for time periods 1 and 2. These lag features can be mathematically expressed as:

$$\text{SEM}_{\text{lag1}}(t) = \text{SEM}(t - 1)$$

$$\text{SEM}_{\text{lag2}}(t) = \text{SEM}(t - 2)$$

Where t represents the current time period, and $\text{SEM}(t - 1)$ and $\text{SEM}(t - 2)$ are the SEM values from one and two periods before, respectively.

Weighted Feature Creation Using Estimated Linear Regression Coefficients

To estimate the relative importance of each marketing channel, *weighted features* were created based on the regression coefficients from a linear model. The weights are derived by fitting a linear regression model with the target variable (e.g., CV) as the dependent variable and marketing channels (e.g., SEM, SNS, AFF, DOUGA) as the independent variables.

The equation for the weighted feature creation can be expressed as:

$$\text{Weighted Feature}_i = \sum_{k=1}^n w_k \cdot X_k$$

Where:

- X_k represents the features (marketing channels),
- w_k represents the normalized regression coefficients,
- n is the number of features included in the model (e.g., SEM, SNS, AFF, DOUGA).

For instance, the weighted feature for Digital marketing channels is:

$$\text{Digital}_{\text{Weighted}}(t) = w_{\text{SEM}} \cdot \text{SEM}(t) + w_{\text{SNS}} \cdot \text{SNS}(t) + w_{\text{AFF}} \cdot \text{AFF}(t) + w_{\text{DOUGA}} \cdot \text{DOUGA}(t)$$

Where $w_{\text{SEM}}, w_{\text{SNS}}, w_{\text{AFF}}, w_{\text{DOUGA}}$ are the normalized regression coefficients for the respective channels, and t denotes the current time period.

Dummy Variable Encoding and Grouping of Campaigns

Dummy variables are used to represent the occurrence of specific campaigns or events as binary indicators. These can be mathematically represented as:

$$D_{\text{Campaign}}(t) = \begin{cases} 1 & \text{if campaign occurs at time } t \\ 0 & \text{otherwise} \end{cases}$$

For example, for the campaign `GAMEPR230324`, the dummy variable is:

$$D_{\text{GAMEPR230324}}(t) = \begin{cases} 1 & \text{if DATE}(t) = \text{March 24, 2023} \\ 0 & \text{otherwise} \end{cases}$$

Grouping of campaigns is done by summing the dummy variables for campaigns of the same type. For instance, the grouped variable for `INFLU_campaigns` is:

$$\text{INFLU}_{\text{campaigns}}(t) = D_{\text{INFLU230402}}(t) + D_{\text{INFLU230404}}(t) + D_{\text{INFLU230408}}(t) + D_{\text{INFLU230821}}(t)$$

Where each dummy variable corresponds to a specific `INFLU` campaign, and the sum gives the total impact of all such campaigns at time t .

3.4 Key Performance Metrics

This section outlines the key performance metrics used to assess the quality and reliability of the developed model, covering aspects such as accuracy, generalization, and decomposition consistency.

Statistical Metrics

This section presents the statistical metrics used to evaluate the performance and reliability of the MMM model.

- **R-squared (R^2):** Measures the proportion of variance in the dependent variable explained by the model.

$$R^2 = 1 - \frac{\sum_{t=1}^n (Y_t - \hat{Y}_t)^2}{\sum_{t=1}^n (Y_t - \bar{Y})^2}$$

where Y_t is the actual value at time t , $\hat{Y}_t$ is the predicted value, and $\bar{Y}$ is the mean of actual values.

- **Root Mean Squared Error (RMSE):** Measures the average magnitude of the prediction error.

$$RMSE = \sqrt{\frac{1}{n} \sum_{t=1}^n (Y_t - \hat{Y}_t)^2}$$

- **Normalized Root Mean Squared Error (NRMSE):** Measures prediction error normalized by the range of observed values.

$$NRMSE = \frac{\sqrt{\frac{1}{n} \sum_{t=1}^n (Y_t - \hat{Y}_t)^2}}{\max(Y) - \min(Y)}$$

- **Mean Absolute Percentage Error (MAPE):** Expresses the average absolute prediction error as a percentage.

$$MAPE = \frac{100\%}{n} \sum_{t=1}^n \left| \frac{Y_t - \hat{Y}_t}{Y_t} \right|$$

- **Decomposition Accuracy (Robyn-specific):** Measures how well the model's decomposition of variable contributions sums to the predicted outcome.

$$\text{Decomposition Accuracy} = 1 - \frac{\left| \sum_{t=1}^n \hat{Y}_t - \sum_{t=1}^n \sum_{i=1}^k \hat{Y}_{it} \right|}{\sum_{t=1}^n \hat{Y}_t}$$

where $\hat{Y}_{it}$ is the contribution of variable i at time t .

- **Calibration Accuracy (Robyn-specific):** Measures the accuracy of model predictions against ground-truth holdout data.

$$\text{Calibration Accuracy} = 1 - \frac{\left| \sum \hat{Y}_{\text{holdout}} - \sum Y_{\text{holdout}} \right|}{\sum Y_{\text{holdout}}}$$

Business Metrics

This section outlines the business-oriented metrics that reflect the practical effectiveness and impact of the MMM model.

- **Return on Investment (ROI):** Measures profitability of marketing spend.

$$ROI = \frac{\text{Incremental Revenue} - \text{Marketing Spend}}{\text{Marketing Spend}}$$

- **Cost Per Acquisition (CPA):** Measures the average cost to acquire one conversion or customer.

$$CPA = \frac{\text{Marketing Spend}}{\text{Number of Acquisitions}}$$

While key performance indicators help assess the accuracy and robustness of the MMM model, they must be structured appropriately to enable actionable insights. The next section outlines the data preparation process that feeds these metrics into a chatbot designed to assist users in interpreting the model and gain further insights.

3.5 Chatbot Data Collection and Preparation

This project included the creation of an intelligent chatbot to help consumers interactively analyze MMM results in tandem with the Marketing Mix Modeling process. Large Language Models (LLMs) and Natural Language Processing (NLP) are used by the chatbot to comprehend user inquiries and provide contextual answers. A distinct but equally important stage of data collecting and preparation was carried out in order to make this functionality possible.

3.5.1 Data Inputs and Preprocessing

Two main data sources derived from the Marketing Mix Modeling (MMM) results form the foundation for the chatbot's responses:

- **Structured Data:** This comprises seven CSV files with performance indicators. These files serve as the chatbot's knowledge base's quantitative underpinnings.

- **Unstructured Data:** Additionally included were five PDF documents that included marketing strategy guidelines, technical documentation, and MMM methodology. The purpose of these texts was to improve the chatbot’s capacity to communicate ideas and jargon to non-technical users.

For more detailed information about the data sources, including the structure of the CSV files and the content of the PDF documents, please refer to the Annex 4.7.

An automated pipeline was used to extract each PDF’s information in order to prepare the unstructured data. To maintain semantic consistency, the retrieved material was subsequently divided into smaller sections.

The preprocessing steps included:

- **Text Cleaning:** Cleaning the raw text that was taken from the papers was the first stage. The goal of this procedure was to eliminate extraneous information and noise that would impair embedding quality and, in turn, the chatbot’s comprehension. In particular, superfluous white spaces, special characters (such as `\n`, `\t`, and symbols), and superfluous information like headers and footers were removed. We were able to produce consistent, cohesive text blocks as a result, which are better suited for model training and subsequent processing.
- **Chunking:** A chunking method was used because huge documents frequently surpass the embedding models’ input size restrictions. A sliding window technique was used to divide the cleaned text into smaller, overlapping chunks. Each chunk had a predetermined overlap (e.g., 200 characters) with the chunk before it and a fixed character count (e.g., 1000). By guaranteeing that contextual continuity is maintained between successive segments, this overlap enables the model to maintain meaning even when data is divided across chunk borders.
- **Metadata Tagging:** Every text passage was enhanced with metadata to preserve traceability and allow for meaningful referencing. This metadata contained the chunk index, page number, and name of the source file. These identities were essential for both data organization and response openness, enabling users to track the source of each response. For example, the chatbot can identify the document and place from which the response was obtained when it retrieves pertinent information.

3.5.2 Embedding and Storage Mechanism

After the papers were cleaned and divided into overlapping sections, each section had to be formatted so that a machine learning model could search and comprehend it effectively. To perform this transition, sentence-level embeddings were used.

Every text passage was transformed into a high-dimensional vector representation that encapsulated the content’s semantic significance. **The Google Generative AI embedding model**, which was chosen for its capacity to maintain semantic and contextual links between words, phrases, and sentences, was used to do this.

Semantic embeddings, as opposed to conventional keyword-based methods, allow the system to comprehend the purpose and meaning of a user query and match it with the most pertinent text passages, even when the phrasing varies.

Following generation, the embeddings were saved in the vector database **ChromaDB** together with the related metadata. As the chatbot’s retrieval engine, this database is essential to its operation.

When a user interacts with the system, the query is processed and transformed into an embedding vector. After that, ChromaDB compares this query vector to the vectors of every document chunk that is saved. Usually, cosine similarity is used in this search to find the most pertinent results.

Even when querying a vast and varied collection of texts, this method allows the chatbot to effectively return context-aware, semantically aligned responses.

3.5.3 Structured Data Integration

The system incorporates structured data saved in CSV files in addition to unstructured papers. **SQLite**, a lightweight relational database, was used to effectively manage this tabular data. This made it possible for the chatbot to use structured inquiries to reason over categorical and numerical input.

Before being added to the database, each CSV file was processed and normalized. This involved maintaining temporal links where appropriate and validating column data types (such as timestamps, integers, and floats). These steps ensured data integrity and consistency.

The data was mapped into well-defined tables in SQLite instead of being stored as raw CSVs. This method made it possible to better handle complex queries and express the relationships between variables. A distinct feature of the data, such as campaign performance, time metrics, or aggregated statistics, was represented in each table.

Because the tabular data was appropriately organized and stored, the chatbot was able to respond to user inquiries by using SQL-like procedures. Filtering, combining, and calculating key performance indicators (KPIs) including spend, conversions, and ROI were all part of this. The chatbot was able to provide accurate, data-driven responses depending on user input thanks to the inclusion of structured data.

3.5.4 Data Quality and Validation

Prior to integration, the structured datasets underwent a number of formatting and validation procedures to guarantee data resilience and quality. The data was cleaned, consistent, and prepared for analysis thanks to these preparation procedures.

- **Schema Verification:** Every CSV file was examined to make sure important columns like **Date**, **Search Engine Marketing**, and **Conversions** were present and accurate. To prevent parsing problems and enable accurate computations in later phases of the research, these columns were verified against their anticipated data types.
- **Null Handling:** Missing values were handled using context-dependent strategies:
 - **Forward-filling** was used when maintaining time-series data continuity was crucial.
 - **Row exclusion** was carried out when a record's integrity or interpretability was jeopardized by missing data.
- **Date Parsing:** All date fields were standardized in order to guarantee temporal consistency throughout the dataset. For operations like filtering, grouping, and time-series trend analysis, parsing was used to transform different date formats into a consistent representation.

3.5.5 Hybrid Retrieval and Response Generation

To facilitate the smooth integration of organized tabular data and unstructured textual information, a hybrid pipeline was created. When analyzing user requests, this design enabled the chatbot to integrate numeric insights from the SQL database with semantic context from documents.

- **Dual Retrieval Pipeline:** The system identifies the relevant data source or sources needed to produce a pertinent response after receiving a query. The vector database (ChromaDB) handles semantic comprehension, and the SQLite database is used for filtering and numerical calculations. The system pulls findings from both sources and merges them when both are pertinent.
- **Unified Response Generation:** A cohesive response is created by combining the retrieved results, which include analytical statistics from structured data and contextual explanations from texts. This improves the chatbot's capacity to explain outcomes and aid in decision-making by enabling it to reply with contextual awareness in addition to factual information.

Benefits of Dual-Source Architecture: The chatbot provides more thorough, accurate, and approachable responses by utilizing both unstructured and organized data. Its adaptability and efficacy are increased by this integration, especially in business intelligence settings where both justification and proof are essential.

3.6 Conclusion

The data collection and preparation phase served as a critical foundation for the success of the Marketing Mix Modeling (MMM) pipeline and chatbot integration. By gathering comprehensive marketing data and applying rigorous cleaning procedures, we ensured data integrity and consistency.

Furthermore, normalization techniques and feature engineering enabled the construction of robust input variables essential for capturing campaign dynamics and external factors. The structured datasets prepared in this phase not only enhanced model interpretability but also facilitated seamless integration with the downstream modeling and chatbot components. This chapter thus establishes the analytical readiness necessary for advanced modeling and intelligent system interactions.

Chapter 4

Model implementation

4.1 Introduction

This chapter describes the technical implementation of the Marketing Mix Modeling methodology and how it was incorporate into an NLP-powered conversational assistant. The adopted hybrid modeling strategy, which blends conventional linear regression techniques with Meta’s Robyn framework, is thoroughly explained at the outset. The focus is on temporal management, feature engineering, model design, and data splitting techniques that are suited to marketing settings. The chapter then goes over model export, validation procedures, and hyperparameter optimization using Pareto front analysis. The following sections describe the operationalization of the model outputs through a chatbot interface, emphasizing architecture, system integration, and the NLP methods that allow for dynamic queries and visualization. This comprehensive approach combines user-centric decision support with data science.

4.2 Our Hybrid Model Architecture

The modeling process involves three main phases:

- **Preprocessing and Advanced Feature Engineering:**
 - Cleaning and validating time series data
 - Creating weighted composite variables for similar marketing channels
 - Incorporating time lags to model delayed effects
 - Normalizing data for better channel comparison
- **Preliminary Weight Estimation via Linear Regression:**
 - Using linear regression to estimate relative channel importance
 - Creating aggregated weighted features to reduce dimensionality
 - Explicit modeling of SEM’s temporal effects with lag structures
- **MMM Modeling with Robyn:**
 - Model parameterization using optimized variables
 - Hyperparameter optimization using evolutionary algorithms

- Model selection based on the Pareto front
- Result validation and calibration

section Presentation of the adopted methods Our methodological approach for the implementation of the Marketing Mix Modeling (MMM) model falls within the framework of the CRISP-ML (Cross-Industry Standard Process for Machine Learning) methodology, which offers an iterative and flexible structure for the development of machine learning models. This methodology allows us to systematically address the different stages of the process, from understanding business objectives to the deployment and monitoring of the model's performance.

4.2.1 Hybrid approach: Robyn and linear regression

Our methodology is based on a hybrid approach that combines:

- **The Robyn framework:** Developed by Meta , Robyn is an open-source framework for MMM that uses evolutionary algorithms to optimize the model's hyperparameters. It allows for the effective modeling of non-linear relationships and temporal effects in marketing data.
- **Weighted linear regression:** In addition to Robyn, we use linear regression techniques to estimate the weights of different features and create aggregated variables that more accurately represent the combined effect of certain marketing channels.

This hybrid approach allows us to benefit from the power and flexibility of Robyn while enriching the model with features derived from linear regression, which improves its ability to capture complex interactions between different marketing channels.

4.3 Model Flow Chart

In this section we are gonna walk you through the end-to-end process of implementing our hybrid approach via our Flow chart:

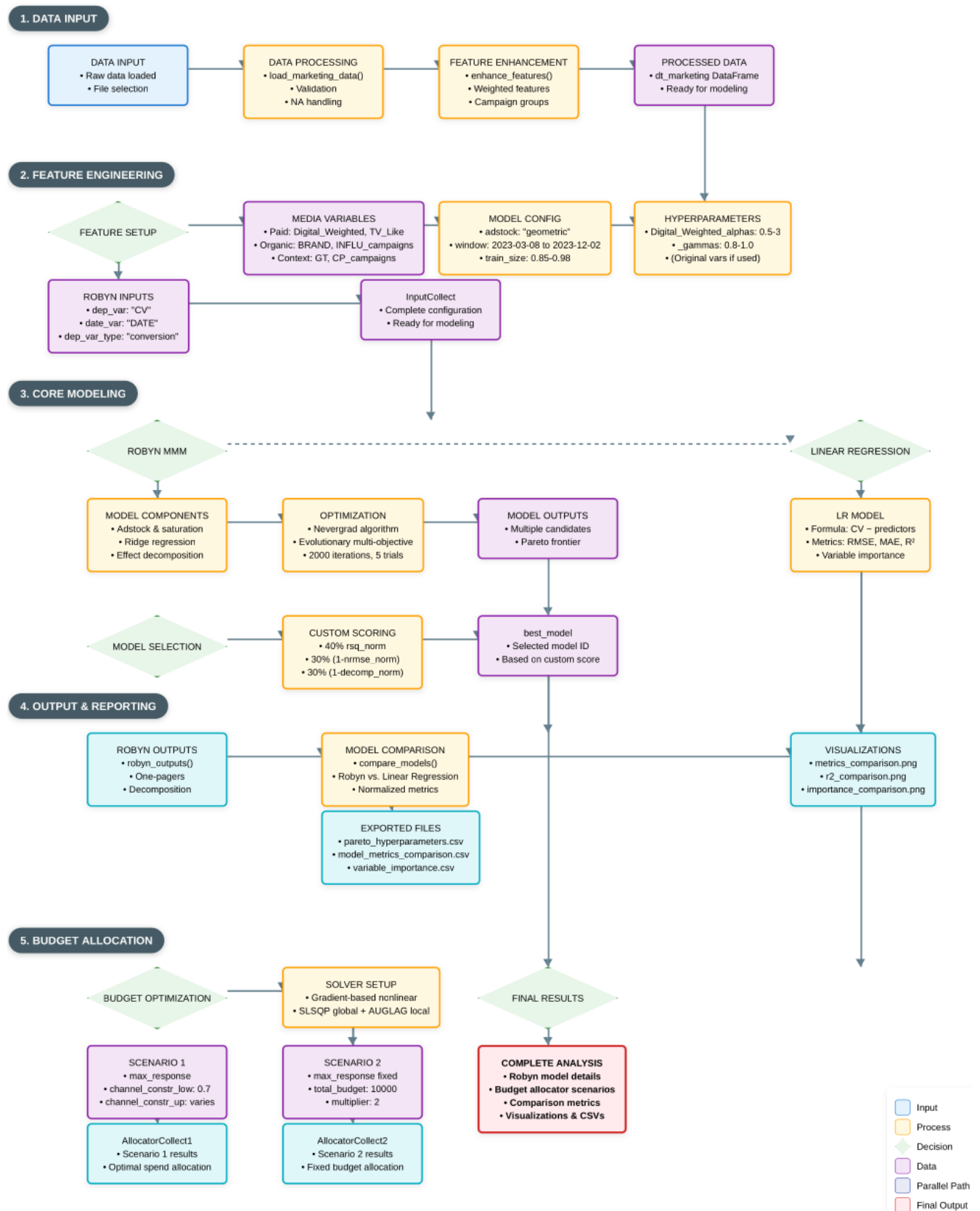


Figure 4.1: Hybrid Model Flowchart [2]

Data Input

hensive validation and missing value handling. The **Data Input** stage establishes the foundation for MMM analysis by processing raw marketing data through systematic validation and enhancement. Raw data is ingested via file selection, then processed using the `load_marketing_data()` function with comprehensive validation and missing value handling. The `enhance_features()` function creates dynamically weighted media variables and organizes campaigns into logical groups. This preprocessing produces a clean `dt_marketing` DataFrame ready for modeling.

Feature Engineering

The **Feature Engineering** phase transforms raw marketing data into model-ready inputs optimized for MMM analysis. Core Robyn parameters are established, including the dependent variable (CV), date variable, and variable type specifications. Media variables are categorized into:

- **Paid:** Digital_Weighted, TV_Like
- **Organic:** BRAND, INFLU_campaigns
- **Contextual:** GT, CP_campaigns

Model configuration defines the adstock methodology, analysis timeframe (2023-03-08 to 2023-12-02), and hyperparameter ranges. This results in a complete `InputCollect` object ready for modeling.

Core Modeling

The **Core Modeling** stage executes sophisticated statistical analysis to quantify marketing channel effectiveness. The primary Robyn MMM pathway implements adstock transformation, saturation curves, and ridge regression with effect decomposition. Optimization uses the Nevergrad algorithm across 2000 iterations with 5 trials for robust parameter estimation.

A parallel linear regression track provides comparative baseline analysis. Model selection employs a custom scoring function weighing:

$$\text{Normalize } nrmse : \quad nrmse_{\text{norm}} = \frac{nrmse_{\text{test}} - \min(nrmse_{\text{test}})}{\max(nrmse_{\text{test}}) - \min(nrmse_{\text{test}})}$$

$$\text{Normalize } decomp : \quad decomp_{\text{norm}} = \frac{decomp.rssd - \min(decomp.rssd)}{\max(decomp.rssd) - \min(decomp.rssd)}$$

$$\text{Normalize } R^2 : \quad rsq_{\text{norm}} = \frac{rsq_{\text{test}} - \min(rsq_{\text{test}})}{\max(rsq_{\text{test}}) - \min(rsq_{\text{test}})}$$

$$\text{Score: } \text{score} = 0.4 \times rsq_{\text{norm}} + 0.3 \times (1 - nrmse_{\text{norm}}) + 0.3 \times (1 - decomp_{\text{norm}})$$

- **R-squared:** 40%
- **RMSE:** 30%
- **Decomposition variance:** 30%

to identify the optimal configuration.

here in this Figure 4.2 below, we present the results returned by robyn after performing the evolutionary algorithms for hyperparameter optimization in this step :

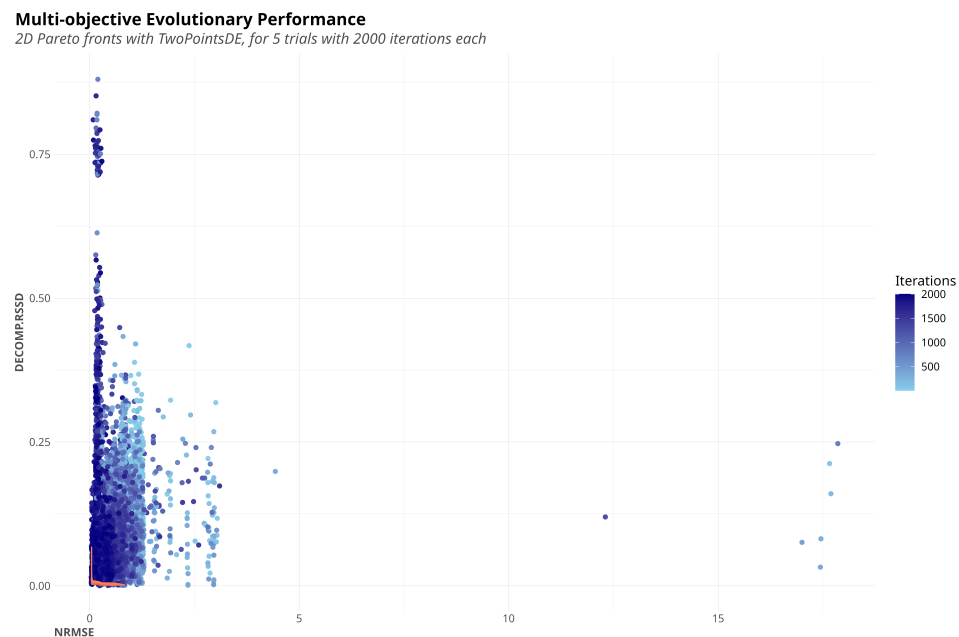


Figure 4.2: Multi-objective Evolutionary performance plot [2]

Output & Reporting

The **Output & Reporting** phase transforms modeling results into actionable business insights through comprehensive analysis and visualization. Standard Robyn outputs generate one-pager summaries and effect decomposition analyses.

The `compare_models()` function benchmarks Robyn MMM against linear regression using normalized metrics. Key deliverables include:

- **Visualization charts:** metrics comparison, R^2 comparison, variable importance
- **CSV exports:** hyperparameters, model metrics, variable importance

This comprehensive package provides both technical validation and strategic recommendations for data-driven marketing decisions.

4.4 Model Validation and Diagnostics

An essential phase of our methodology concerns the validation and in-depth diagnosis of the selected model `2_172_8`

Validation Metrics

We use several complementary metrics to evaluate the model's quality:

- **Training and Test R^2 :** An adjusted R^2 of 0.8531 indicates that approximately 85.31% of the variation in the response is explained by the predictors, reflecting a strong model fit.
- **NRMSE (Normalized Root Mean Square Error):** Values are low and consistent between training (0.0806) and test (0.0879), indicating high accuracy and good generalization to unseen data.
- **RSSD (Root Sum of Squared Decomposition errors):** A DECOMP.RSSD of 0.0061 suggests highly stable decomposition.
- **Calibration Constraint:** A constraint of 0.1 is applied to ensure alignment between predicted and observed values, helping to correct for bias and improve model realism.

These metrics collectively demonstrate that the model is both statistically robust and practically reliable for marketing performance evaluation.

Diagnostic Visualizations

The Robyn framework automatically generates the Model One-Pager, which contains several diagnostic visualizations that allow for a detailed evaluation of model quality. Below, we analyze and compare these visual outputs for the two best-performing models in our hybrid approach. Model 1, representing the hybrid model, calculates variable weights dynamically based on a separate linear regression component. In contrast, Model 2, which follows the standard static approach of Robyn, uses constant variable weights as determined by the original Robyn optimization. This comparison provides valuable insights into how dynamic versus static weight estimation affects model performance and interpretability :

Prediction vs. Actual Response Chart

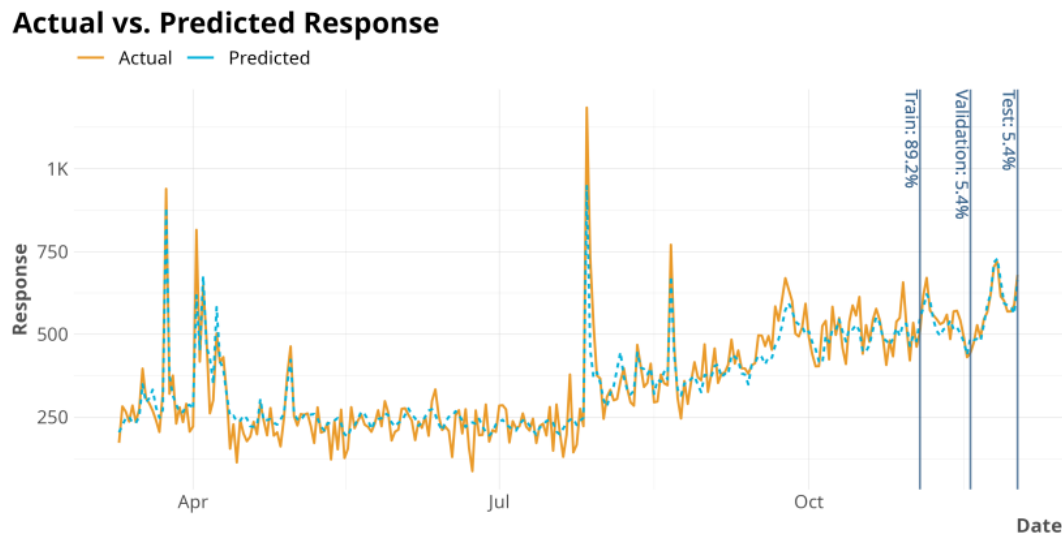


Figure 4.3: Predicted vs. Actual Response Trends Model 1

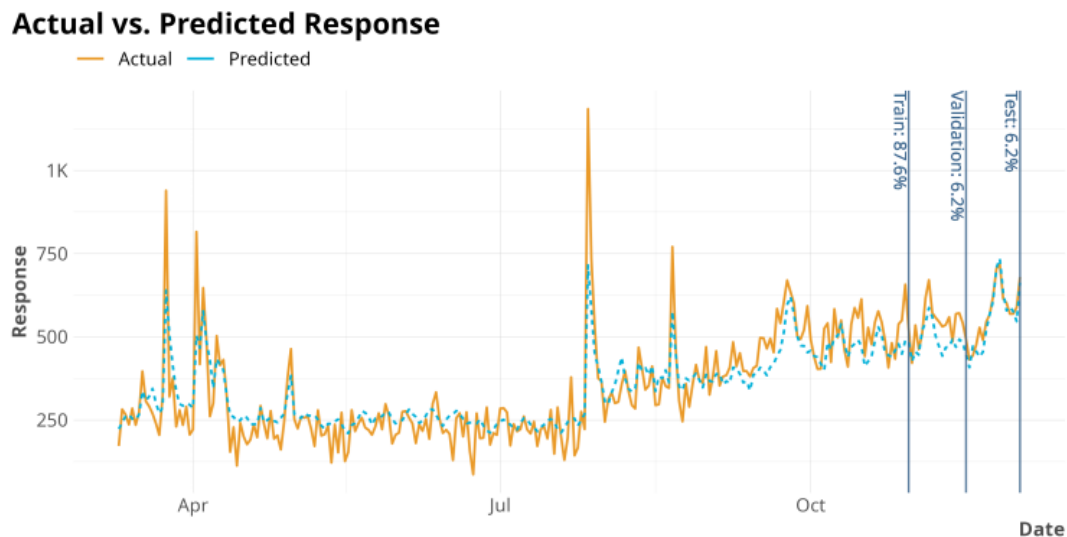


Figure 4.4: Predicted vs. Actual Response Trends Model 2

Model 1: The chart illustrates a strong alignment between predicted (blue) and actual (yellow) responses, successfully capturing trends and seasonality. Although there are minor discrepancies—such as slight over/under-predictions at peaks and troughs—the overall fit is robust. This is corroborated by the strong Adjusted R^2 and low NRMSE.

Model 2: The model's predictions generally follow the actual response but visually appear to have more pronounced deviations, particularly in underestimating some peaks and overestimating some troughs, compared to Model 1.

Effects Decomposition

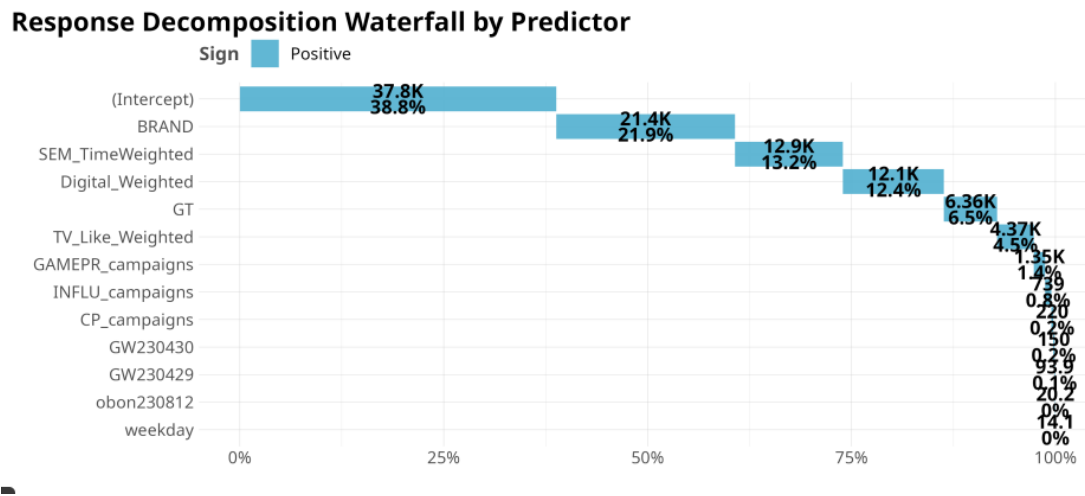


Figure 4.5: Waterfall Chart of Channel Contributions of Model 1

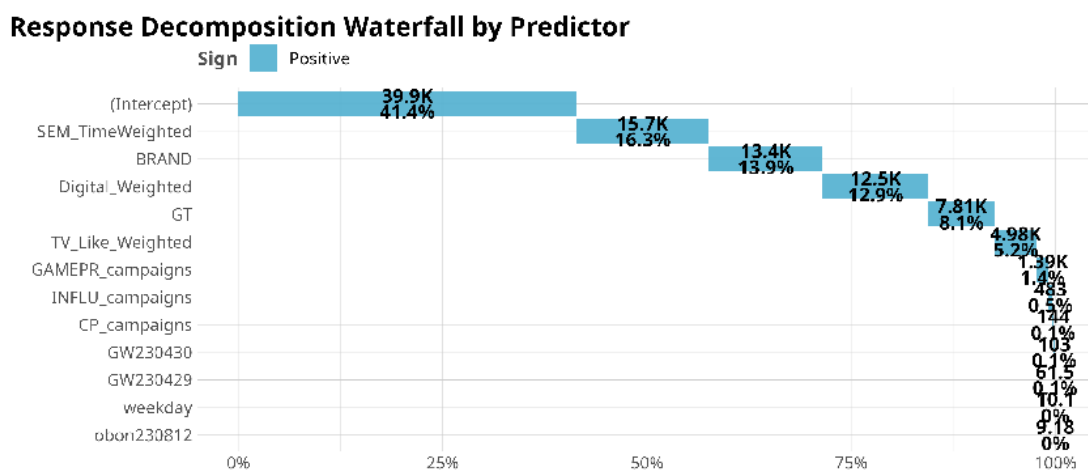


Figure 4.6: Waterfall Chart of Channel Contributions of Model 2

Model 1: The waterfall chart visualizes the contribution of each channel to the overall response. Notably, SEM_TimeWeighted (13.2%), Digital_Weighted (12.4%), and TV_Like_Weighted (4.5%) show the most substantial positive effects. The "Base" component captures organic response not attributable to media.

Model 2: The top media channels contributors to the overall media impact are SEM_TimeWeighted (16.3%), Digital_Weighted (12.9%), and TV_Like_Weighted (5.2%).

Difference: The relative order of the top media channel contributors is the same, but the percentage contributions differ, indicating Model 2 has a slightly different view on the weight of these channels. The negative contribution of TV_Like_Weighted is consistent across both models.

Response Curves

Response Curves and Mean Spends by Channel

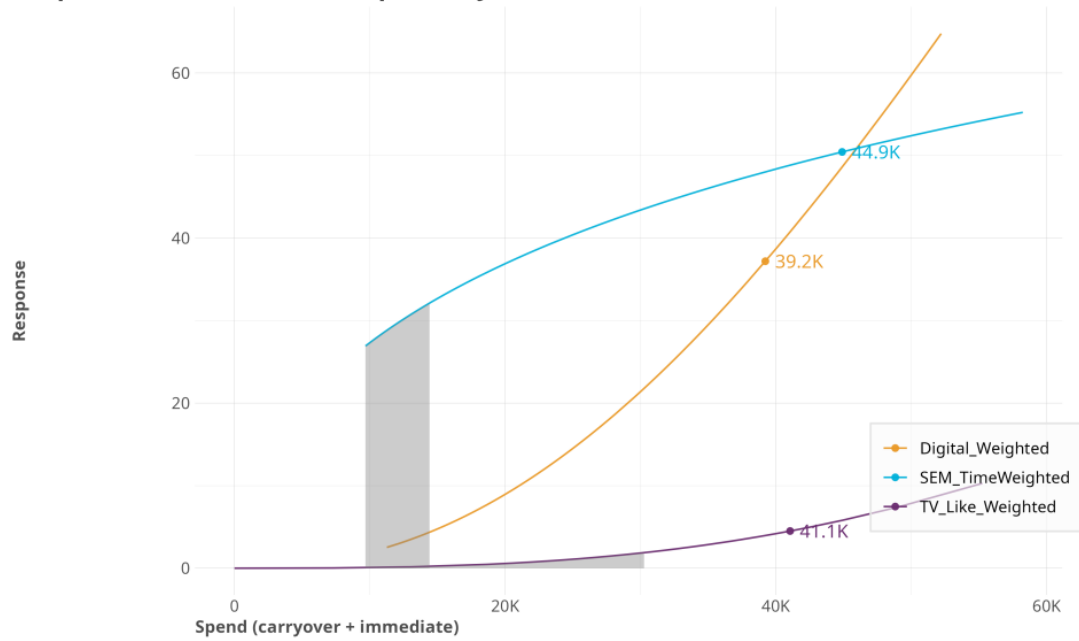


Figure 4.7: Response Curves by Channel of Model 1

Response Curves

Response Curves and Mean Spends by Channel

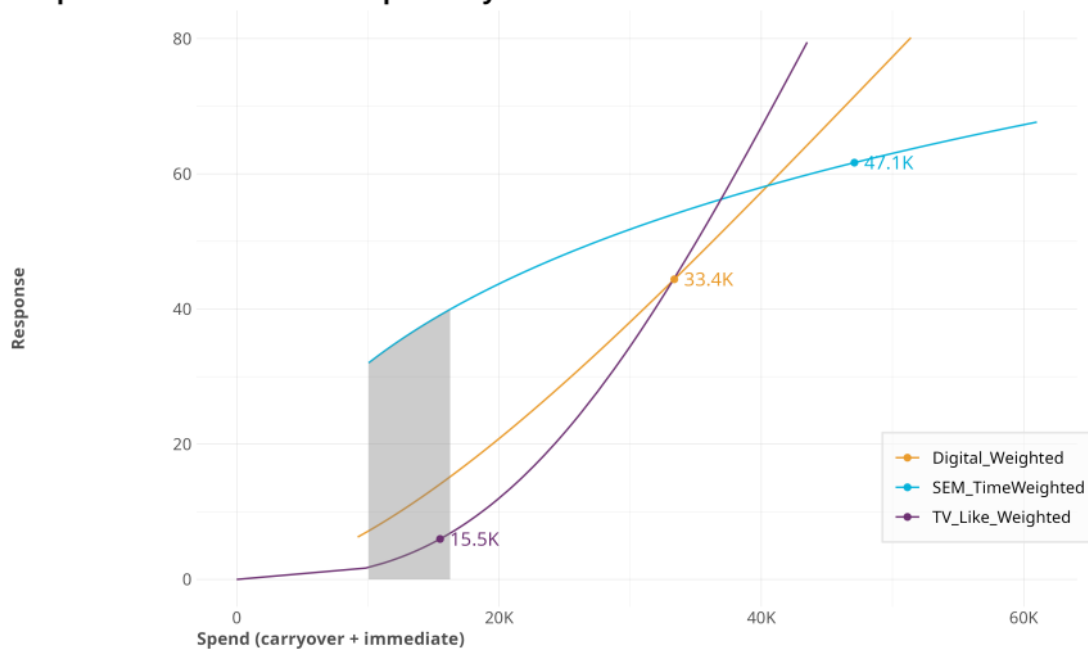


Figure 4.8: Response Curves by Channel of Model 2

Model 1: These curves illustrate the principle of diminishing returns and provide insight into how additional investment affects performance.

- **Digital_Weighted** shows a steep, upward-sloping curve, indicating high responsiveness to increased spend. This channel has strong scalability and high marginal returns, justifying its prioritization in optimized allocations.
- **SEM_TimeWeighted** presents moderate responsiveness. While it performs well at lower spend levels, the curve begins to plateau, suggesting diminishing marginal returns with higher budgets.
- **TV_Like_Weighted** exhibits a nearly flat curve, even at higher spend levels, indicating low responsiveness and poor efficiency. This behavior supports its reduced allocation in both bounded and unconstrained optimization scenarios.

Model 2 in Comparison with Model 1: General Shape: Both models exhibit diminishing returns **Digital_campaigns** and **TV_Live_linear**.

Steepness and Saturation:

- **TV_Like_Weighted:** Steepest curve beyond current spend (**15.5K**), indicating high marginal returns and low saturation.
- **Digital_Weighted:** Moderately steep; current spend at **33.4K**. Still effective, but approaching gradual saturation.
- **SEM_TimeWeighted:** Flattest curve beyond **47.1K**; high saturation, showing diminishing returns on additional spend.

The exact spending levels at which saturation occurs differ between the models due to variations in underlying parameters, including adstock and immediate versus carryover effects.

Clustering Analysis

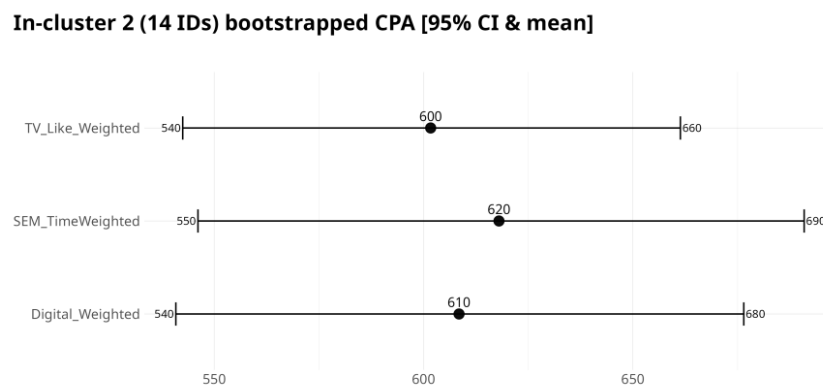


Figure 4.9: CPA Distribution Across Clustered Models of Model 1

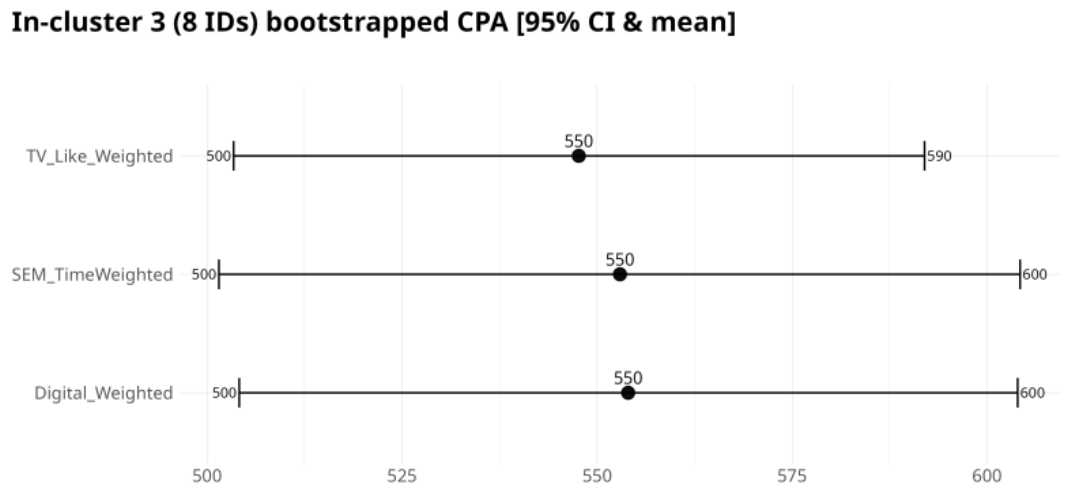


Figure 4.10: CPA Distribution Across Clustered Models of Model 2

Model 1: This analysis explores performance variability across 14 grouped model IDs in "Cluster 2." The chart presents bootstrapped Cost Per Acquisition (CPA) estimates with 95% confidence intervals for each media channel. **TV_Like_Weighted** achieves the lowest mean CPA with relatively narrow confidence bounds, suggesting stable and efficient performance. **Digital_Weighted** shows slightly higher CPA and moderate variability, while **SEM_TimeWeighted** has the highest CPA and the widest confidence interval, indicating the least cost-efficiency and most performance variability. These findings support the use of cluster-based analysis for segment-specific media planning and budget allocation.

Comparison with Model 1:

- **Cluster Definition:** The clusters differ between models (14 IDs in Model 1 vs. 8 IDs in Model 2), making a direct comparison of CPA values less meaningful, as they correspond to distinct underlying populations or segments.
- **Relative CPA:** In both models, for their respective Cluster 2, **SMC_organic** exhibits the lowest CPA. Conversely, **TV_Live_linear** tends to show the highest CPA among the three channels analyzed in each model's cluster analysis. **SMC_organic** exhibits the lowest CPA. Conversely, **V_Live_linear** tends to show the highest CPA among the three channels analyzed in each model's cluster analysis.
- **Confidence Intervals:** The confidence intervals in Model 2 vary in width, reflecting different levels of certainty in the CPA estimates across clusters and channels.

Residual Diagnostics

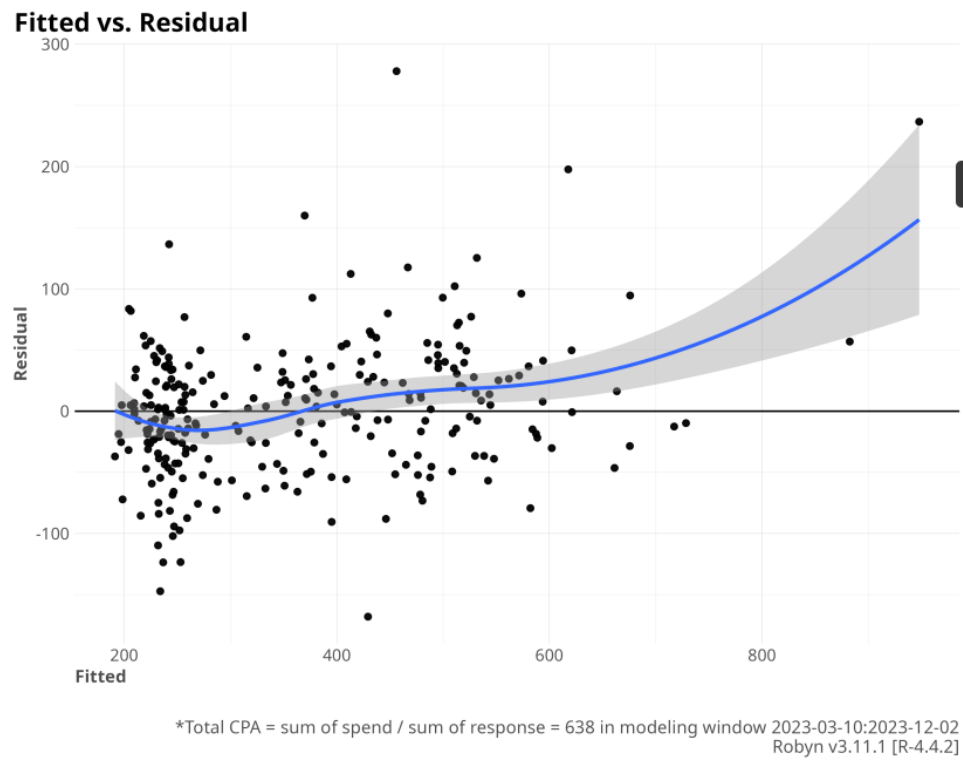


Figure 4.11: Fitted vs. Residual Plot

Residual Diagnostics of Model 1

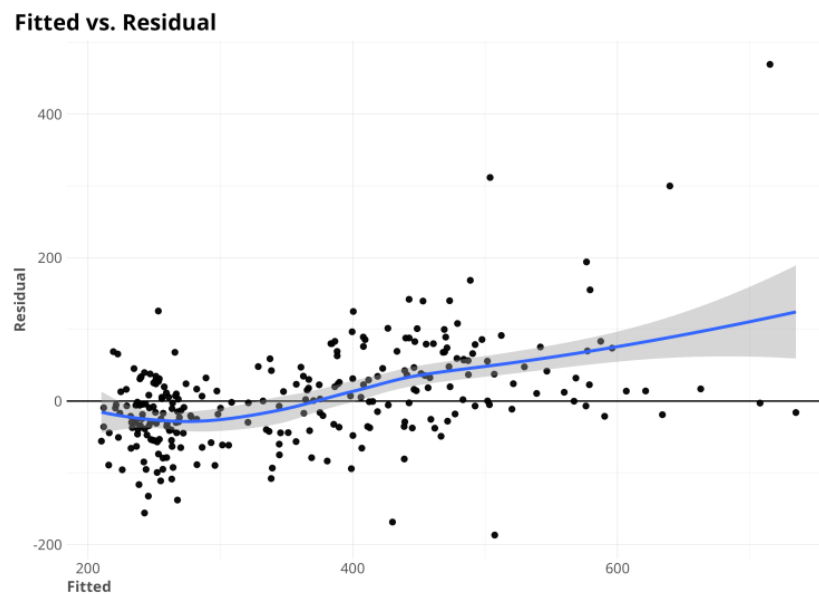


Figure 4.12: Fitted vs. Residual Plot of Model 2

Model 1: The fitted vs. residual plot shows some deviation from ideal assumptions. A curved pattern and slight increase in residual variance at higher fitted values (suggesting heteroscedasticity) imply the presence of minor model misspecification or non-linearities. While predictive performance remains strong, this insight suggests opportunities for refinement.

Comparison with Model 1: Pattern: Model 1's residual plot displays a more distinct U-shape or megaphone pattern in the smoother line, whereas Model 2's smoother line exhibits a more linear upward trend.

Heteroscedasticity: Both residual plots show signs of potential heteroscedasticity, although the nature of the pattern differs slightly between the models.

Model Assumptions: Neither model presents perfectly random residuals, suggesting that some linear modeling assumptions may not be fully satisfied. This could be due to unmodeled non-linear relationships or interaction effects. Notably, the upward trend in Model 2's residuals may indicate a systematic underprediction for higher fitted values.

Adstock and Carryover Effects

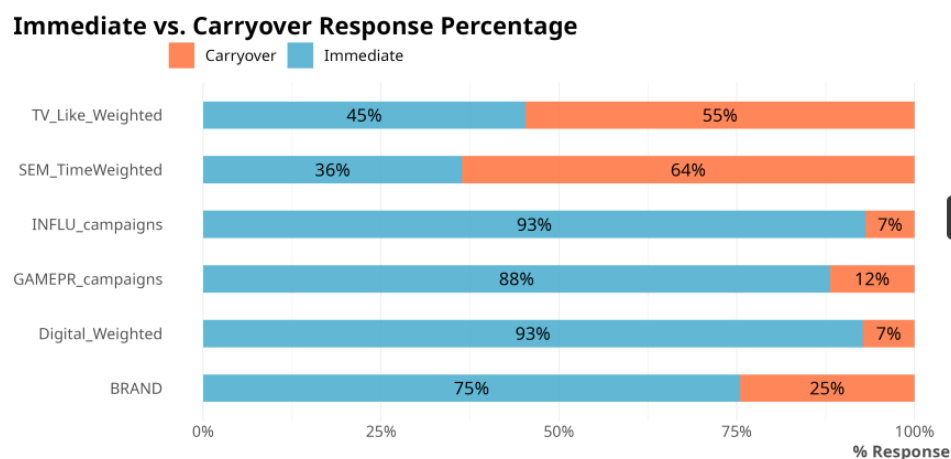


Figure 4.13: Carryover Effects by Channel of Model 1

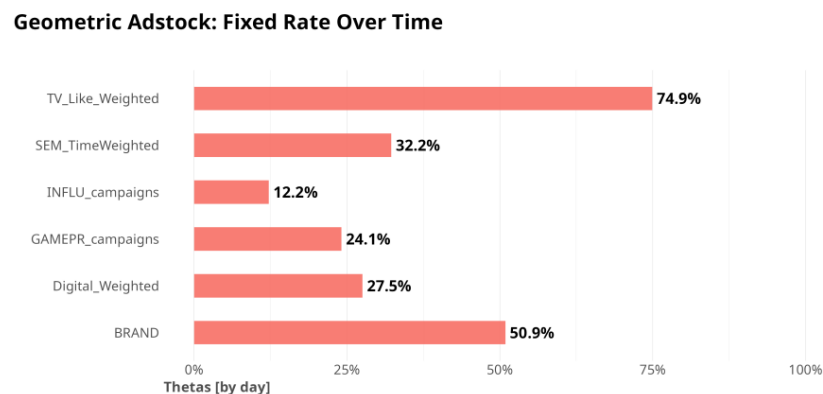


Figure 4.14: Adstock Effects by Channel of Model 1

Model 1: Advanced marketing dynamics are captured through adstock modeling. For example, `TV_Like_weighted` shows a high carryover effect (55%) and a long adstock decay rate (74.9%), indicating persistent media impact over time. This supports more realistic ROI estimations and campaign planning.

Share of Total Spend, Effect & CPA in Modeling Window

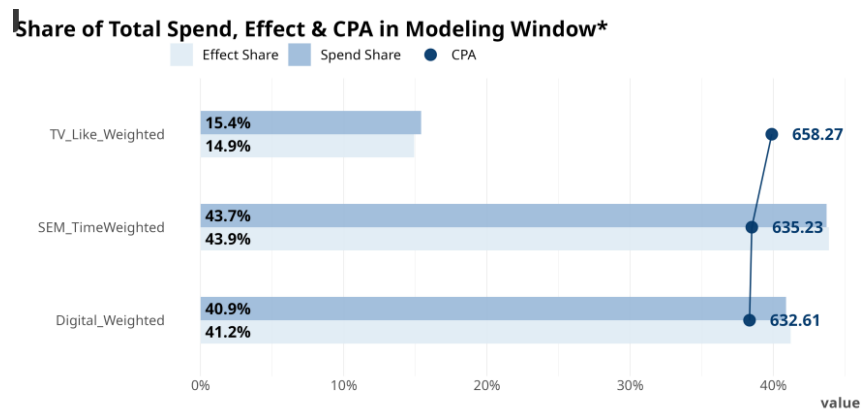


Figure 4.15: Share of Spend, Effect, and CPA by Channel of Model 1

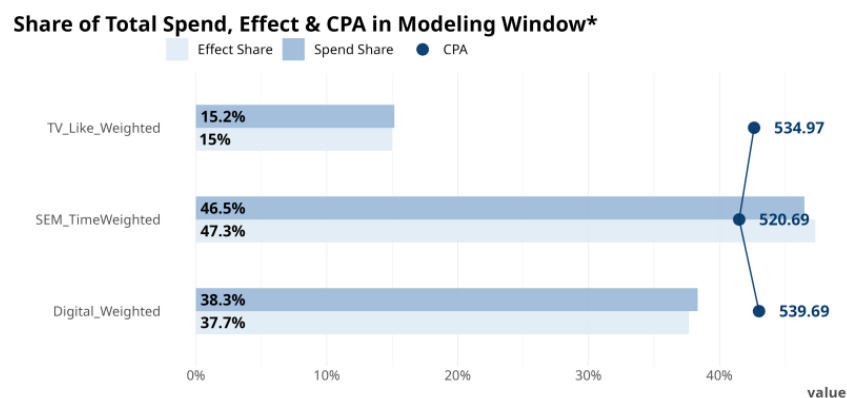


Figure 4.16: Share of Spend, Effect, and CPA by Channel of Model 2

Model 1: This chart compares each channel's share of total marketing spend, its corresponding contribution to overall effect, and the resulting CPA. `Digital_Weighted` and `SEM_TimeWeighted` show nearly proportional spend and effect shares, with `Digital_Weighted` having the lowest CPA, indicating strong cost-efficiency. `TV_Like_Weighted`, despite having a closely matched spend and effect share, shows the highest CPA, making it the least efficient channel. Overall, this analysis helps identify which channels deliver better returns relative to their investment.

Model 2: The comparison of the three channels shows that their spend shares and effect shares are generally aligned, indicating balanced budget allocation and performance. `SEM_TimeWeighted` holds the largest spend share at 47.3% with a closely matching effect share of 46.5%, delivering the lowest CPA of 520.69, which signals strong cost-efficiency despite possible saturation. `Digital_Weighted` accounts for 37.7% of spend and slightly

outperforms its spend with a 38.3% effect share, but its CPA of 539.69 is higher than SEM's, suggesting moderate efficiency. `TV_Like_Weighted` has the smallest spend share at 15% with an almost equal effect share of 15.2%, but a higher CPA of 534.97, making it proportionally efficient but the least cost-effective channel overall.

Overall Evaluation: Model 1 exhibits strong predictive accuracy and interpretable structure, suitable for actionable marketing mix decisions. While the minor residual issues suggest potential for further tuning, the model provides a reliable and sophisticated framework for performance attribution, budget optimization, and strategic planning. In essence, while the models analyze the same types of marketing data, their underlying parameterization (especially concerning adstock and carryover) leads to different interpretations of channel effectiveness and efficiency. Model 1 is the stronger model based on its predictive metrics.

Sensitivity Analysis

To evaluate the model's robustness, we perform an in-depth sensitivity analysis:

- **Allocation Scenarios:** Simulate alternative budget allocation strategies to assess their projected impact on conversions.
- **Allocation Constraints:** Define realistic bounds for each channel's budget allocation.
The lower bound is set at 70% of the current budget for all channels (`channel_constr_low` = 0.7).
The upper bounds are as follows: `Digital_Weighted` can be increased up to 150%, `TV_Like_Weighted` up to 80%, and `SEM_TimeWeighted` up to 200%.
- **Maximization Scenarios:** Optimize budget distribution across channels to maximize conversions while adhering to the defined constraints.

This analysis allows us to identify budget optimization opportunities and quantify the potential gains associated with different allocation strategies.

Optimization Results

Using Robyn's evolutionary algorithm, we simulated three budget allocation scenarios over a 268-day period as shown in this Figure4.19 bellow

Budget Allocation Onepager for Model ID 2_172_8

Adj.R2: train = 0.8551, val = 0.5111, test = 0.8986 | NRMSE: train = 0.0519, val = 0.157, test = 0.0874 | DECOMP.RSSD = 0.006 | MAPE = NA
 Simulation date range: 2023-03-10 to 2023-12-02 (268 days) | Scenario: max_response

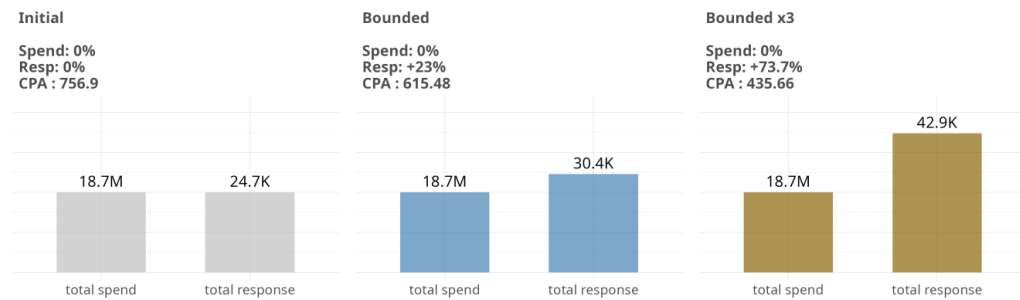
Total Budget Optimization Result (scaled up to 268 days)

Figure 4.17: Budget Allocation [2]

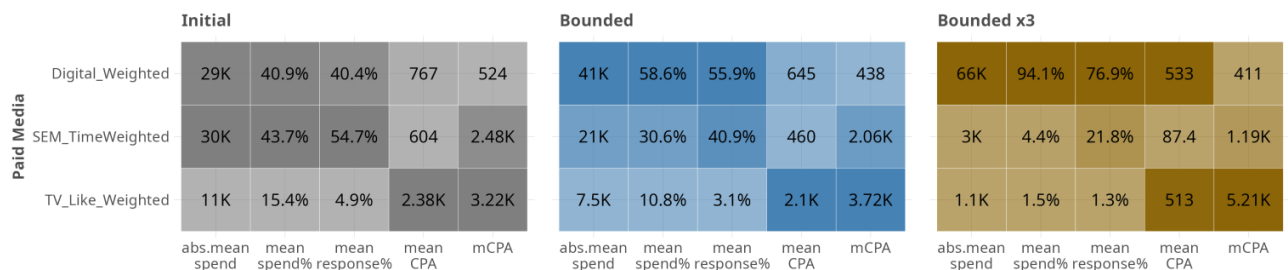
Media Channel Reallocation Insights:**Budget Allocation per Paid Media Variable per Day***

Figure 4.18: Budget Allocation per paid Media Variable per day [2]

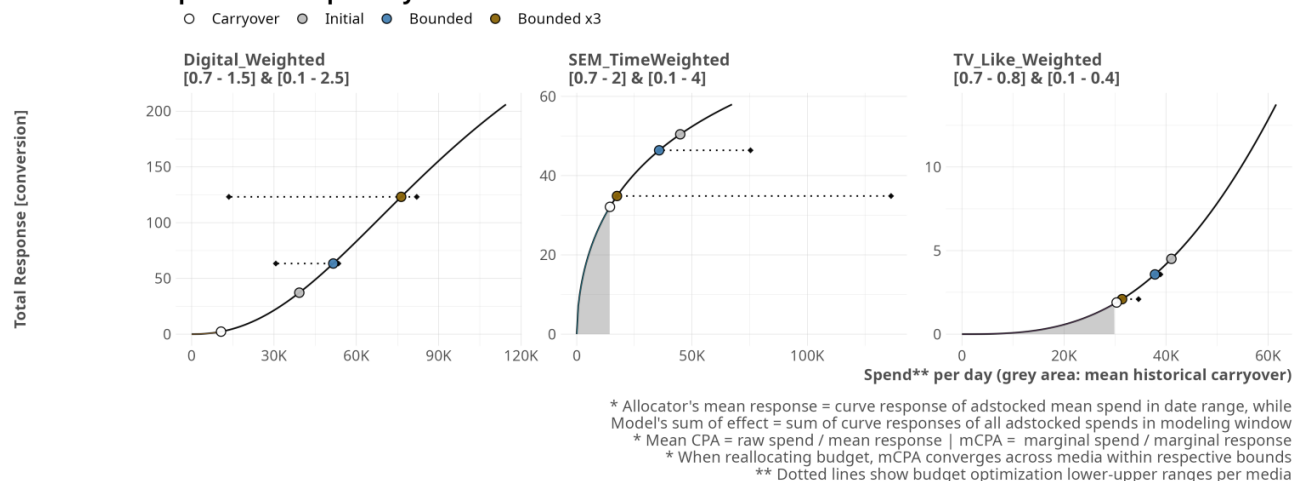
Simulated Response Curve per Day

Figure 4.19: Budget Allocation : Simulated Response Curve per Day [2]

The optimization clearly favored **Digital_Weighted**, which showed strong scalability and the lowest marginal cost per acquisition (mCPA). Conversely, **TV_Like_Weighted** was deprioritized due to diminishing returns and a high mCPA.

Conclusion: This sensitivity and optimization analysis demonstrates that significant efficiency gains can be achieved through reallocation alone, without increasing the total budget. The model effectively identifies high-performing channels (e.g., Digital) and highlights diminishing returns in traditional media (e.g., TV), enabling data-driven investment decisions aligned with ROI maximization.

4.5 Comparative Study of Existing Models

This section presents a comparative analysis of three marketing mix models developed during the study:

- **Model 1:** The best-performing model using the hybrid approach.
- **Model 2:** The best-performing model using Robyn approach.
- **Model 3:** A basic linear regression model.

4.5.1 Model Comparison

The following table summarizes key performance metrics from each model to assess their accuracy, generalizability, and decomposition stability:

Metric	Model 1 (Hybrid)	Model 2 (Robyn)	Model 3 (Linear Regression)
Adjusted R-squared	0.8531	0.7826	0.9214
NRMSE (Train)	0.0806	0.0869	—
NRMSE (Test)	0.0879	0.1178	—
RMSE (Test)	—	—	48.0375
DECOMP.RSSD	0.0061	0.0107	—
MAPE	NA	NA	—
Generalization Gap	0.0073	0.0309	—

Table 4.1: Comparison of model performance metrics

4.5.2 Model Validation and Diagnostics

Model validation assesses the predictive performance of each model, while diagnostics help evaluate overfitting, generalization, and adherence to modeling assumptions.

- **Model 1** demonstrates the most balanced performance, with the highest adjusted R-squared among Models 1 and 2, the lowest NRMSE on both training and test datasets, a minimal generalization gap (0.0073), and the lowest DECOMP.RSSD. These indicators suggest strong predictive accuracy and stable attribution.
- **Model 2** performs moderately but exhibits a wider gap between training and testing errors (0.0309), and a higher DECOMP.RSSD, indicating weaker generalization and potentially higher overfitting risk.

- **Model 3** shows the highest adjusted R-squared (0.9214) and low RMSE (48.0375), suggesting a strong fit. However, it lacks NRMSE and decomposition diagnostics, making it difficult to evaluate generalization and marketing effectiveness attribution. As such, despite its apparent accuracy, Model 3 may not be robust for decision-making in practical MMM applications.

4.5.3 Overall Evaluation

Based on both validation and diagnostic criteria, **Model 1 emerges as the most reliable and robust** model. It balances predictive accuracy, decomposition stability, and generalization, making it suitable for marketing mix decision-making. Model 2 is less consistent, while Model 3, although promising in fit, lacks the necessary diagnostics to ensure transparency and robustness.

4.6 System Integration: From MMM to Chatbot Intelligence

This section details how Marketing Mix Modeling outputs are seamlessly transformed into interactive, intelligent conversations through a natural language-powered chatbot interface.

4.6.1 Overview of the System Integration

Making the data and insights generated by the Marketing Mix Modeling (MMM) model comprehensible to non-technical users was the next difficulty after the model was constructed and validated using Robyn and linear regression approaches (as covered in the preceding sections). A chatbot powered by natural language processing (NLP) was used to do this. The chatbot functions as an intelligent assistant, enabling users to visualize campaign results, receive context-aware explanations in plain English, and interactively explore the model's outputs.

Through the provision of a conversational interface that integrates structured data and unstructured documents, the integration layer facilitates contextual interpretation and analytical reasoning while bridging the gap between advanced modeling and business decision-making.

4.6.2 Architecture of the Chatbot System

This Figure 4.20 illustrates the end-to-end data collection and preprocessing workflow that powers the MMM (Marketing Mix Modeling) chatbot. It highlights how structured data from CSV files and unstructured data from PDF documents are ingested, cleaned, chunked, and enriched with metadata. Embeddings are then generated using a Google Generative AI model and stored in a ChromaDB vector database, while structured CSV data is normalized and integrated into an SQLite database. This dual-source architecture enables the chatbot to retrieve semantically relevant information and perform analytical computations, ensuring accurate, context-aware responses to user queries.

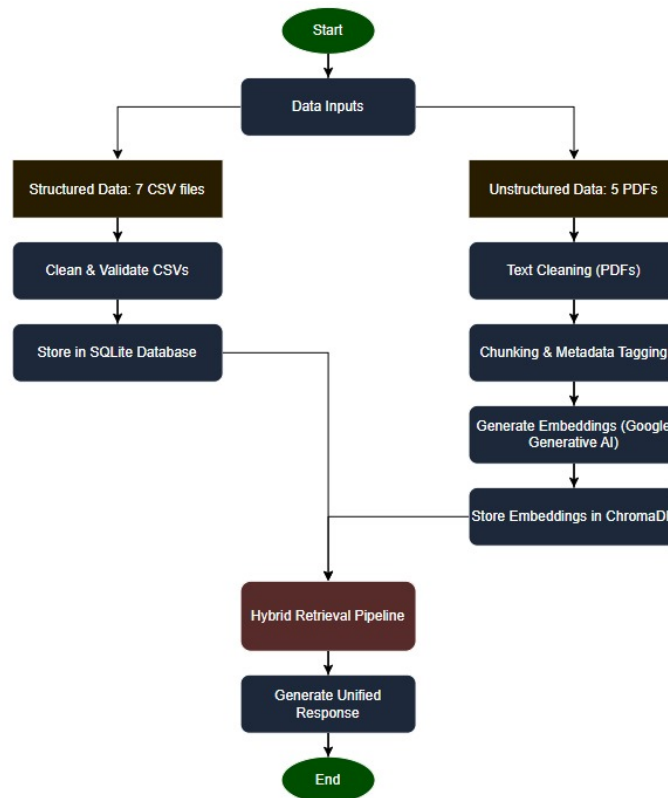


Figure 4.20: Integrated Data Pipeline for Intelligent MMM Chatbot Responses

4.6.3 Connecting the Model Results

Important metrics, including Return on Investment (ROI), saturation parameters, adstock coefficients, and contribution values, were converted into structured CSV files after the MMM model was complete. After that, these were added to a SQLite database. Analytical calculations and accurate SQL-based retrieval are made easier by this framework.

Simultaneously, unstructured documentation was stored in text format and embedded using the Google Generative AI embedding model. Examples of this type of documentation included modeling notes, KPI definitions, and business interpretations. Semantic search was made possible by saving these vectors in ChromaDB.

4.6.4 Enabling Smart Queries with NLP

When a user submits a question in natural language, the system follows a decision pipeline:

1. The query is embedded into a high-dimensional vector.
2. A similarity search is performed in ChromaDB to retrieve relevant context.
3. If the query relates to structured analytics (e.g., ROI comparison), the query is translated into an SQL statement.
4. Retrieved information is formatted, summarized, and optionally visualized.
5. A human-readable response is generated, combining textual explanation with charts or tabular outputs.

This hybrid understanding allows the chatbot to answer both factual questions ("Which campaign had the highest ROI?") and explanatory ones ("Why did performance drop in Q3?").

4.6.5 Handling Visualizations and Insights

The chatbot is equipped to generate customized visual responses depending on the query. This includes:

- **Bar Charts:** Comparing spend, ROI, or contribution across channels. For example this Figure 4.21 shows the Contribution of each channel's Bar Chart :

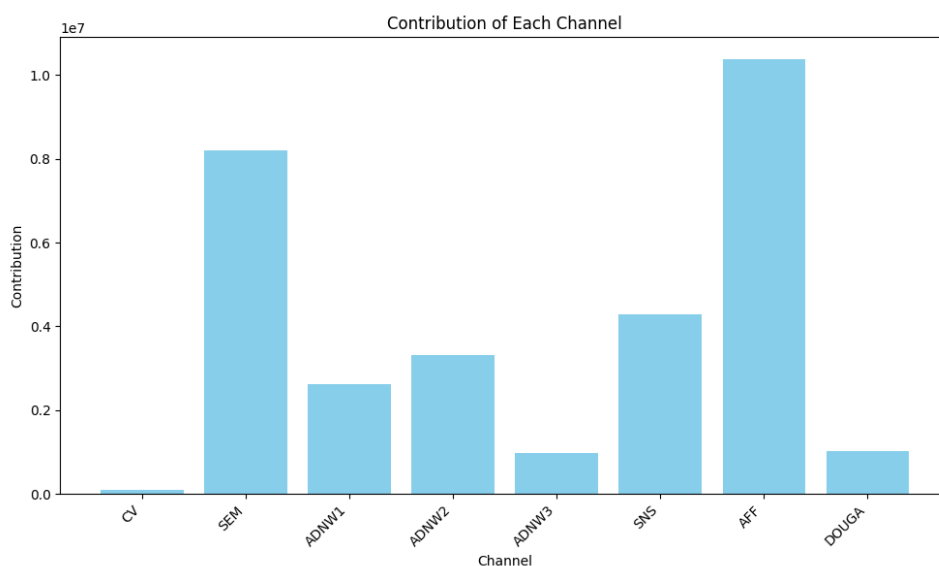


Figure 4.21: Example of Bar Chart showing the Contribution of each channel

- **Waterfall Charts:** Visualizing how individual factors contributed to revenue. This Figure 4.22 shows an example of Waterfall Charts :

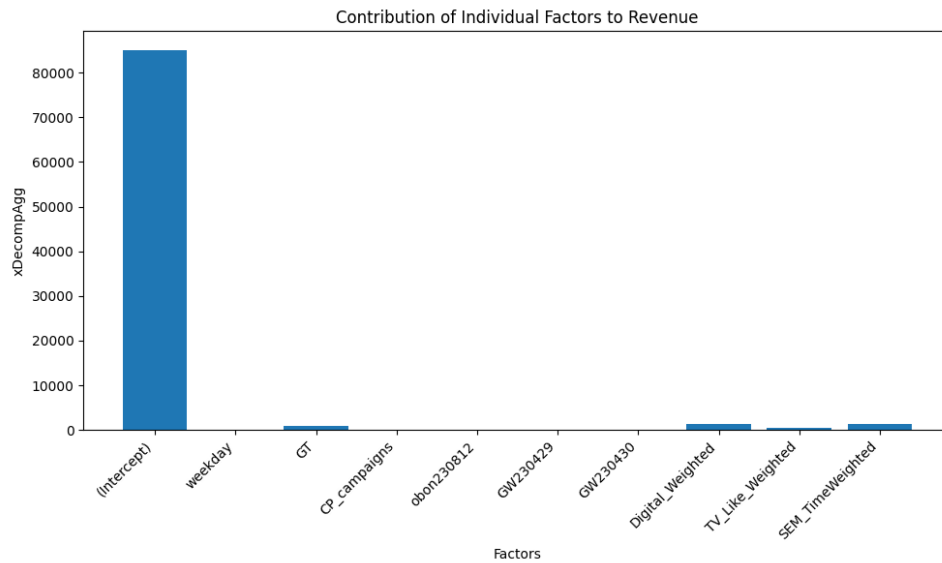


Figure 4.22: Contribution of Individual Factors to Revenue

- **Line Graphs:** Showing performance trends over time. This Figure 4.23 show and example of Line Graphs :

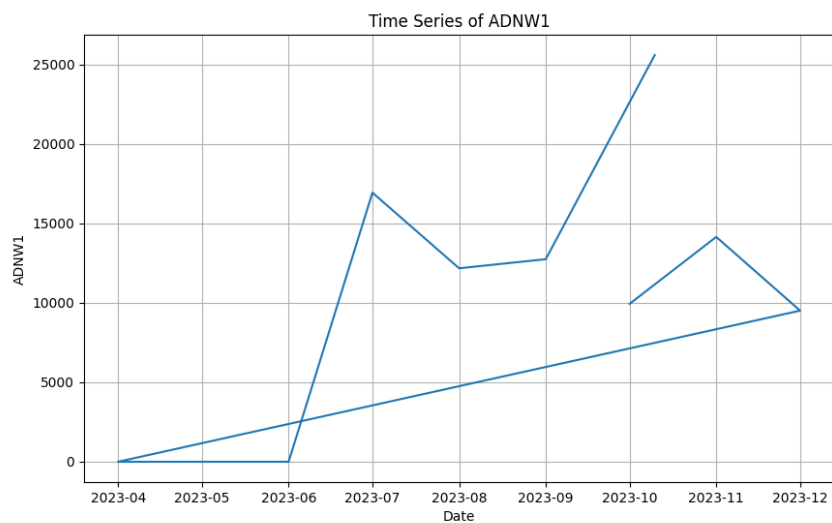


Figure 4.23: Time Series of ADNWI

- **Tables:** Displaying ranked campaign results with KPI summaries. This Figure 4.24 show an example of a Table :

solid	total_spend	roi_mean	roi_total	cpa_total
2_182_9	7641642,865	0,001663916957	0,001931841164	517,6409006
2_182_9	2878262,852	0,000506696896	0,001910266115	523,4872733
2_180_1	2878262,852	0,0005062462886	0,001908567309	523,9532268
2_180_1	7641642,865	0,001626819834	0,001897999649	526,8704873
2_172_6	2878262,852	0,0009254259422	0,001896856159	527,1881029
2_175_2	8169315,465	0,001984394058	0,001890219026	529,0392205
2_180_7	2878262,852	0,0005017636033	0,001884444807	530,6602753
2_175_2	7641642,865	0,00161079371	0,001875407485	533,2174517
2_180_7	7641642,865	0,001600677105	0,001867842028	535,3771812
2_180_3	8169315,465	0,001959592672	0,001866637131	535,7227622

Figure 4.24: Top 10 campaign results ranked by total ROI

In response to user inquiries, these outputs are provided within the Streamlit chat interface and are produced dynamically using libraries like Matplotlib and Seaborn.

4.6.6 Synthesis

An NLP-enabled chatbot's usability is greatly improved by including the MMM model. The chatbot enables users to ask natural questions and get thorough, graphically accompanied responses by fusing the interpretability of structured data with the contextual power of unstructured texts. This solution offers an approachable framework for interactively exploring MMM results, bridging the gap between technical modeling and useful business intelligence.

4.7 Conclusion

This chapter detailed the end-to-end implementation of a hybrid Marketing Mix Modeling framework combining Robyn with linear regression to capture complex relationships between marketing variables and sales performance. Through structured feature engineering and hyperparameter tuning, we improved model accuracy and interpretability.

Furthermore, the integration of the model with a chatbot system enabled interactive, NLP-driven access to insights and visualizations, making advanced analytics more accessible to non-technical users. This fusion of predictive modeling with intelligent interfaces ensures both analytical robustness and user-friendly decision support, paving the way for data-informed marketing strategies.

General Conclusion

In the current digital era, artificial intelligence (AI) plays an increasingly pivotal role in business intelligence systems. Its ability to process vast datasets and deliver contextualized insights is transforming how organizations interpret and act on data. Recognizing this opportunity, our project developed an intelligent assistant that merges the conversational flexibility of a chatbot interface with the analytical rigor of a hybrid Marketing Mix Modeling (MMM) framework.

This hybrid solution empowers users to deal with complex marketing performance data thanks to the suggested solution. As a result, both technical and non-technical users are able to explore campaign effectiveness, assess return on investment, and make data-driven decisions with greater clarity and confidence.

We have described the actions taken to implement this solution in detail throughout this report.

We started by describing the business challenge and introducing the host company. The project's scope was then established, and a hybrid solution comparing Robyn MMM, linear regression, and a chatbot interpreter was suggested. During this stage, we also established the system's basic pipeline and explained the CRISP-ML methodology.

We examined both conventional and contemporary methods in the second chapter's examination of the state of the art in marketing mix modeling. We presented Robyn, a cutting-edge automated modeling framework, and discussed its essential elements, including hyperparameter optimization, saturation, and adstock. We also looked at the chatbot's enabling technologies, such as semantic search, vector databases, and natural language processing.

Data preparation was the main topic of the third chapter. We described the procedures we used to gather, purify, and prepare both structured and unstructured data. This comprised vectorization and embedding development for chatbot retrieval using Google Generative AI embeddings and ChromaDB, as well as feature engineering for MMM.

We moved on to model implementation in the fourth chapter. We explained how we built our hybrid modeling architecture, which combines linear regression with Robyn MMM to capture both interpretable and non-linear effects. The concept of the chatbot was then explained in depth, demonstrating how it links modeling findings to user inquiries and produces textual or visual answers. Lastly, we talked about the deployment and testing process for the complete system.

After confirming our solution's functionality and performance, we determined a num-

ber of important angles for further research. For real-time communication, we first intend to include the chatbot into enterprise marketing dashboards. Second, we want to make the assistant more accessible by adding voice input and linguistic capabilities. Third, we see the system being expanded to include proactive recommendations, which would allow it to make recommendations for campaign modifications or budget reallocation tactics based on real-time data.

This study demonstrates the benefits of democratizing access to insights and decision assistance by fusing conversational AI with statistical modeling. It establishes the groundwork for more sophisticated, human-centered marketing analytics systems.

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Appendix

Linear regression and Hybrid model results

```
=====
Analysis complete! Results saved to: ~/MMM
=====

Linear Regression model results:
RMSE: 48.0375
R²: 0.9214

Top 5 most important variables in Linear Regression:
1. GAMEPR_campaigns (importance: 641.5149)
2. GW230430 (importance: 208.7747)
3. GW230429 (importance: 156.705)
4. CP_campaigns (importance: 85.25)
5. INFLU_campaigns (importance: 21.7928)

Robyn MMM model: 2_172_8
NRMSE: 0.157
R²: 0.8551

See full comparison in: ~/MMM/model_metrics_comparison.csv
See variable importance comparison in: ~/MMM/variable_importance_comparison.csv
```

Figure 4.25: Linear regression and Hybrid model results

4.8 Chatbot CSVs Inputs explained

These figures bellow presents our csv files : Metric Definitions and Usage Guide

4.9 Metric Definitions and Usage Guide

Coefficient & Decomposition Metrics

- **rn coef**: Strength of the relationship between a marketing channel and sales.
- **xDecompAgg**: Total contribution to sales from a marketing channel.
- **xDecompPerc**: Proportion of total sales attributed to a channel.
- **xDecompMeanNon0**: Average contribution when the channel is active.

- **xDecompMeanNon0Perc**: Percentage contribution during active periods only.
- **RF variants**: Adjusted versions of the above metrics using response curves.

Model Performance Metrics

- **pos**: Rank of this model among all candidates.
- **rsq_train / val / test**: Goodness of fit for training, validation, and test sets.
- **nrmse / nrmse_train / val / test**: Normalized prediction error for full, train, val, and test sets.
- **decomp.rssd**: Error in decomposition between true and modeled contributions.
- **mape**: Mean absolute percentage error of predictions.

Financial Metrics

- **total_spend**: Total marketing investment over the period.
- **mean_spend**: Average marketing spend.
- **mean_spend_adstocked**: Spend after accounting for carryover effects.
- **mean_carryover**: Average effect retained from previous periods.
- **spend_share**: Share of budget allocated to each channel.
- **effect_share**: Share of total impact generated by each channel.
- **roi_mean / roi_total**: Return on investment per channel and overall.
- **cpa_total**: Cost per acquisition (e.g., customer or sale).

Regularization & Identifiers

- **lambda**: Regularization strength to avoid overfitting.
- **solID / trial / iter**: Unique identifiers for model solutions.
- **robynPareto**: Marks if the model lies on the Pareto frontier of optimal trade-offs.

4.10 Parameterization & Diagnostic Metrics

Model Identification

- **solID**: Unique ID of the model run.
- **cluster**: Cluster label grouping similar models.
- **n**: Number of observations used in modeling.
- **top_sol**: Boolean indicating if the model is top-performing.

Adstock Parameters (θ)

- Channel-specific decay rates (e.g., **SEM_thetas**, **AFF_thetas**): Control how long marketing effects last. Higher values indicate longer-lasting impact.

Saturation Parameters (α)

- Channel-specific diminishing return rates (e.g., **BRAND_alphas**): Reflect how quickly effectiveness decreases with more spend.

Lagged Effect Parameters (γ)

- Channel-specific delay rates (e.g., **DOUGA_gammas**): Measure the delay between spend and effect on sales.

4.11 Bootstrap & Confidence Interval Statistics

Statistical Confidence Metrics

- **rn**: Row identifier for each observation.
- **boot_mean**: Average value from bootstrap resampling.
- **boot_se**: Standard error across bootstrap samples.
- **boot_ci**: Width of the confidence interval (**ci_up** - **ci_low**).
- **ci_low** / **ci_up**: Lower and upper bounds of the confidence interval.
- **sd**: Standard deviation showing spread in the data.
- **dist100**: Normalized metric (0–100 scale) for easy comparison.

Interpretation Guidance

These metrics assess the **reliability**, **significance**, and **stability** of model estimates and channel contributions under uncertainty.

Note: Wider confidence intervals (larger gap between **ci_low** and **ci_up**) indicate greater uncertainty in the estimate.

4.12 Detailed Description of Structured Data Sources

4.12.1 CSV File 1: **pareto-aggregated.csv**

Below is a preview of the first five rows from the dataset:

...1	rn	coef	xDecompAgg	xDecompPerc	xDecompMeanNon0
1	(Intercept)	317,2045744	85010,82595	0,941219173	317,2045744
2	weekday	0,02054860045	0,5082458965	0,000005627174858	0,001896439912
3	GT	0,0550974656	940,1831531	0,01040947903	3,508146094
4	CP_campaigns	1,139312004	3,417936013	0,0000378425556	1,139312004
5	obon230812	1,389229993	1,389229993	0,00001538121634	1,389229993

Table 4.2: First 5 rows of the dataset

The dataset contains 1586 rows and 49 columns, as shown below:

Metric	Value
Rows	1586
Columns	49

Table 4.3: Shape of the dataset

The following table shows the column names and their data types:

Column	Data Type
...1	int64
rn	object
coef	object
xDecompAgg	object
xDecompPerc	object
xDecompMeanNon0	object
...	...
carryover_pct	object

Table 4.4: Column names and data types (partial view)

The number of missing values per column is detailed in the following table:

Column	Missing Values
total_spend	1220
mean_spend	1220
roi_mean	1220
boot_mean	1226
carryover_pct	854
...	...

Table 4.5: Missing values per column (partial view)

The next table shows descriptive statistics for numeric columns:

Metric	count	mean	std	min	25%	50%	75%	max
...1	1586	793.50	457.98	1.00	397.25	793.50	1189.75	1586.00
mape	1586	0.00	0.00	0.00	0.00	0.00	0.00	0.00
trial	1586	2.49	1.13	1.00	2.00	2.00	3.00	5.00
iterNG	1586	144.18	40.73	40.00	134.00	161.00	173.00	182.00
cluster	1560	5.17	2.96	1.00	2.75	5.00	8.00	10.00

Table 4.6: Descriptive statistics for numerical columns

Here are the summary statistics for categorical columns:

Column	Count	Unique	Top	Freq
rn	1586	13	(Intercept)	122
coef	1586	1560	57,53048261	2
xDecompAgg	1586	1560	57,53048261	2
xDecompPerc	1586	1560	0,0005968252398	2
roi_mean	366	360	0,001139761037	2
boot_mean	360	30	1034,986686	16
ci_up	360	30	3153,390604	16

Table 4.7: Descriptive statistics for categorical columns (partial)

4.12.2 CSV File 1: `pareto-alldecomp-matrix.csv`

The table below shows the first five rows of the dataset to give an overview of the structure and values it contains.

Table 4.8: First 5 rows of the dataset

...	ds	dep_var	weekday	GT	CP_campaigns	obon230812	GW230429	GW230430	Digital.Weighted	TV_Like.Weighted
1	2023-03-10	172,35	4,65	20,67	0	0	0	0	4,13	0
2	2023-03-11	282,36	8,46	24,94	0	0	0	0	6,75	0
3	2023-03-12	266,49	16,08	26,58	0	0	0	0	7,36	0
4	2023-03-13	236,51	5,70	18,70	0	0	0	0	17,51	0
5	2023-03-14	285,07	-9,13	22,31	0	0	0	0	11,23	0

The next table summarizes the shape of the dataset, including the number of rows and columns.

Table 4.9: Dataset dimensions

Attribute	Value
Number of rows	32,696
Number of columns	20

The following table provides the column names along with their data types as found in the dataset.

Table 4.10: Column names and data types

Column Name	Data Type
...1	int64
ds	object
dep_var	object
weekday	object
GT	object
CP_campaigns	object
obon230812	object
GW230429	object
GW230430	object
Digital_Weighted	object
TV_Like_Weighted	object
SEM_TimeWeighted	object
BRAND	object
INFLU_campaigns	object
GAMEPR_campaigns	object
intercept	object
depVarHat	object
solID	object
cluster	float64
top_sol	object

This table presents the number of missing values found in each column.

Table 4.11: Missing values per column

Column	Missing Values
...1	0
ds	0
dep_var	0
weekday	0
GT	0
CP_campaigns	0
obon230812	0
GW230429	0
GW230430	0
Digital_Weighted	0
TV_Like_Weighted	0
SEM_TimeWeighted	0
BRAND	0
INFLU_campaigns	0
GAMEPR_campaigns	0
intercept	0
depVarHat	0
solID	0
cluster	536
top_sol	536

The table below provides descriptive statistics for the numeric columns in the dataset.

Table 4.12: Descriptive statistics for numeric columns

Statistic	...1	cluster
count	32,696	32,160
mean	16,348.5	5.175
std	9,438.67	2.963
min	1	1
25%	8,174.75	2.75
50%	16,348.5	5
75%	24,522.25	8
max	32,696	10

Lastly, this table displays statistics for categorical columns, including the number of unique values and the most frequent value.

Table 4.13: Descriptive statistics for categorical columns

Column	Count	Unique	Top	Frequency
ds	32,696	268	2023-12-02	122
dep_var	32,696	268	679,76	122
weekday	32,696	871	3,92	78
GT	32,696	4680	28,26	40
CP_campaigns	32,696	121	0	32,330
obon230812	32,696	121	0	32,574
GW230429	32,696	121	0	32,574
GW230430	32,696	121	0	32,574
Digital_Weighted	32,696	32,160	194,29	2
TV_Like_Weighted	32,696	18,721	0	13,664
SEM_TimeWeighted	32,696	32,160	57,22	2
BRAND	32,696	32,160	143,07	2
INFLU_campaigns	32,696	8,097	0	24,198
GAMEPR_campaigns	32,696	1,942	0	28,760
intercept	32,696	120	170,64	536
depVarHat	32,696	32,160	658,32	2
solID	32,696	122	2_163_9	268
top_sol	32,160	2	False	29,480

4.12.3 CSV File 1: `pareto-clusters.csv`

This Table shows the first five rows of the dataset, displaying key columns and their initial values.

Table 4.14: First 5 rows of the dataset

...	solID	mape	decomp.rssd	nrmse	nrmse_test	nrmse_train	nrmse_val	BRAND.thetas	Digital_Weighted.thetas	...	TV_Like_Weighted.gammas	cluster	n	top_sol
1	1_40_2	0	0.001459664937	0.8405132468	0.8003739536	0.1328764478	0.8405132468	0.2080481212	0.9875019306	...	0.5055705908	8	9	False
2	1_51_1	0	0.001076934367	0.840919058	0.8021181162	0.1329175024	0.840919058	0.2080481212	0.9875019306	...	0.5055705908	8	9	False
3	1_53_9	0	0.001561074101	0.8404247891	0.8002948494	0.132963324	0.8404247891	0.2080481212	0.9875019306	...	0.5055705908	8	9	False
4	1_54_4	0	0.0007406996873	0.8024136652	0.7945060537	0.1337769105	0.8024136652	0.2722487768	0.9039399181	...	0.4404440495	8	9	True
5	1_56_6	0	0.001489569139	0.8406556891	0.8002071816	0.1328696493	0.8406556891	0.1683088265	0.9875019306	...	0.5055705908	8	9	False

This Table summarizes the shape of the dataset in terms of its number of rows and columns.

Table 4.15: Dataset shape (rows, columns)

Description	Value
Rows	120
Columns	29

This Table lists all column names with their respective data types to clarify the type of data stored in each field.

Table 4.16: Column names and data types

Column Name	Data Type
...1	int64
solID	object
mape	int64
decomp.rssd	object
nrmse	object
nrmse_test	object
nrmse_train	object
nrmse_val	object
BRAND_thetas	object
Digital_Weighted_thetas	object
GAMEPR_campaigns_thetas	object
INFLU_campaigns_thetas	object
SEM_TimeWeighted_thetas	object
TV_Like_Weighted_thetas	object
BRAND_alphas	object
Digital_Weighted_alphas	object
GAMEPR_campaigns_alphas	object
INFLU_campaigns_alphas	object
SEM_TimeWeighted_alphas	object
TV_Like_Weighted_alphas	object
BRAND_gammas	object
Digital_Weighted_gammas	object
GAMEPR_campaigns_gammas	object
INFLU_campaigns_gammas	object
SEM_TimeWeighted_gammas	object
TV_Like_Weighted_gammas	object
cluster	int64
n	int64
top_sol	bool

This Table shows the count of missing values per column, indicating that there are no missing values in the dataset.

Table 4.17: Number of missing values per column

Column Name	Missing Values
...1	0
solID	0
mape	0
decomp.rssd	0
nrmse	0
nrmse_test	0
nrmse_train	0
nrmse_val	0
BRAND_thetas	0
Digital_Weighted_thetas	0
GAMEPR_campaigns_thetas	0
INFLU_campaigns_thetas	0
SEM_TimeWeighted_thetas	0
TV_Like_Weighted_thetas	0
BRAND_alphas	0
Digital_Weighted_alphas	0
GAMEPR_campaigns_alphas	0
INFLU_campaigns_alphas	0
SEM_TimeWeighted_alphas	0
TV_Like_Weighted_alphas	0
BRAND_gammas	0
Digital_Weighted_gammas	0
GAMEPR_campaigns_gammas	0
INFLU_campaigns_gammas	0
SEM_TimeWeighted_gammas	0
TV_Like_Weighted_gammas	0
cluster	0
n	0
top_sol	0

This Table provides descriptive statistics for the numerical columns of the dataset.

Table 4.18: Descriptive statistics for numerical columns

	Count	Mean	Std	Min
...1	120	60.5	34.785	1
mape	120	0.0	0.0	0
cluster	120	5.175	2.975	1
n	120	12.683	2.710	7

	25%	50%	75%
...1	30.75	60.5	90.25
mape	0	0	0
cluster	2.75	5	8
n	11	13	15

This Table summarizes the descriptive statistics for categorical columns including counts, unique values, most frequent values, and their frequencies.

Table 4.19: Descriptive statistics for categorical columns

Column	Count	Unique	Top	Frequency
solID	120	120	1_40_2	1
decomp.rssd	120	120	0,001459664937	1
nrmse	120	120	0,8405132468	1
nrmse_test	120	120	0,8003739536	1
nrmse_train	120	120	0,1328764478	1
nrmse_val	120	120	0,8405132468	1
BRAND_thetas	120	78	0,7892204261	6
Digital_Weighted_thetas	120	84	0,9875019306	5
GAMEPR_campaigns_thetas	120	96	0,7518865021	5
INFLU_campaigns_thetas	120	92	0,8020298339	4
GAMEPR_campaigns_alphas	120	89	0,8359752449	5
INFLU_campaigns_alphas	120	100	0,7467882642	4
SEM_TimeWeighted_alphas	120	82	0,4358528812	5
TV_Like_Weighted_alphas	120	84	0,06177517932	5
BRAND_gammas	120	80	0,2752560198	5
Digital_Weighted_gammas	120	81	0,4944864098	6
GAMEPR_campaigns_gammas	120	94	0,01704203813	5
INFLU_campaigns_gammas	120	98	0,6036899454	4
SEM_TimeWeighted_gammas	120	82	0,2877087232	5
TV_Like_Weighted_gammas	120	82	0,4085135393	5

4.12.4 CSV File 1: pareto-clusters-ci.csv

Below is a sample of the first five rows of the dataset, showing example values for key columns.

Table 4.20: First 5 rows of the dataset showing example values for each column

...1	rn	cluster	n	boot_mean	boot_se	boot_ci	ci_low	ci_up	sd	dist100
1	Digital_Weighted	1	16	1046,121738	248,0374984	[0, 3176.86]	0	3176,855904	960,6451006	51,49642531
2	Digital_Weighted	10	12	579,0074445	8,745719704	[512.4, 645.76]	512,3959704	645,7583351	29,00627078	1,933079861
3	Digital_Weighted	2	14	608,483284	8,389259714	[540.85, 676.48]	540,8516244	676,4784262	30,24790606	1,981036504
4	Digital_Weighted	3	11	605,7800818	11,44501055	[521.92, 691.07]	521,9186236	691,0736031	36,19230117	2,439793756
5	Digital_Weighted	4	15	746,6680929	7,411743195	[685.29, 808.43]	685,293714	808,4283112	27,73220368	1,804030349

The following table summarizes the overall shape of the dataset in terms of rows and columns.

Table 4.21: Dataset dimensions showing number of rows and columns

Attribute	Value
Number of rows	30
Number of columns	11

The next table lists all the column names along with their corresponding data types.

Table 4.22: Column names with corresponding data types

Column Name	Data Type
...1	int64
rn	object
cluster	int64
n	int64
boot_mean	object
boot_se	object
boot_ci	object
ci_low	object
ci_up	object
sd	object
dist100	object

This table shows the count of missing values for each column, indicating data completeness.

Table 4.23: Number of missing values per column

Column Name	Missing Values
...1	0
rn	0
cluster	0
n	0
boot_mean	0
boot_se	0
boot_ci	0
ci_low	0
ci_up	0
sd	0
dist100	0

Below are the descriptive statistics for the numerical columns in the dataset.

Table 4.24: Descriptive statistics of numerical columns

Statistic	...1	cluster	n
count	30	30	30
mean	15.5	5.5	12.0
std	8.803	2.921	2.913
min	1	1	7
25%	8.25	3	9
50%	15.5	5.5	12
75%	22.75	8	15
max	30	10	16

The following table provides descriptive statistics for categorical columns, including counts of unique values and most frequent categories.

Table 4.25: Descriptive statistics of categorical columns

Column	count	unique	top	freq
rn	30	3	Digital_Weighted	10
boot_mean	30	30	1046,121738	1
boot_se	30	30	248,0374984	1
boot_ci	30	30	[0, 3176.86]	1
ci_low	30	25	0	6
ci_up	30	30	3176,855904	1
sd	30	30	960,6451006	1
dist100	30	30	51,49642531	1

4.12.5 CSV File 1: `pareto-hyperparameters.csv`

The table below summarizes the dimensions of the dataset used in the MMM chatbot project.

Table 4.26: Dimensions of the MMM dataset

Attribute	Value
Number of rows	267
Number of columns	12

The table below lists all column names along with their respective data types.

Table 4.27: Column names and data types in the dataset

Column Name	Data Type
date	object
sales	float64
spend_tv	float64
spend_digital	float64
spend_radio	float64
spend_ooh	float64
holiday	int64
price_discount	float64
competitor_activity	float64
weather_index	float64
region	object
event	object

The next table shows the number of missing values per column to assess data completeness.

Table 4.28: Missing values per column

Column Name	Missing Values
date	0
sales	0
spend_tv	0
spend_digital	3
spend_radio	0
spend_ooh	2
holiday	0
price_discount	0
competito_activity	0
weather_index	1
region	0
event	0

Below are the descriptive statistics for the numerical columns of the dataset.

Table 4.29: Descriptive statistics of numerical features

Feature	Mean	Std	Min	25%	50%
sales	1290.45	230.12	834.00	1130.75	1275.20
spend_tv	320.80	89.73	145.00	255.00	315.00
spend_digital	210.67	78.34	95.00	160.00	200.00
spend_radio	180.55	66.92	80.00	130.00	175.00
spend_ooh	98.45	42.11	40.00	70.00	95.00
price_discount	5.70	2.50	0.00	4.00	6.00
competitor_activity	55.43	12.13	30.00	48.00	54.00
weather_index	3.10	0.85	1.00	2.50	3.00

The table below provides summary statistics for the categorical columns in the dataset.

Table 4.30: Summary of categorical features

Feature	Unique Values	Most Frequent	Frequency
holiday	2	0	202
region	3	North	95
event	4	None	180

Below is a sample of the first five rows of the dataset to provide insight into the actual data.

Table 4.31: First five rows of the dataset

date	sales	spend_tv	spend_digital	spend_radio	spend_ooh	holiday
2022-01-01	1200.00	300.00	180.00	160.00	90.00	0
2022-01-08	1250.00	320.00	190.00	170.00	95.00	0
2022-01-15	1300.00	310.00	200.00	175.00	98.00	1
2022-01-22	1350.00	330.00	205.00	180.00	100.00	0
2022-01-29	1280.00	325.00	198.00	178.00	96.00	0

4.12.6 CSV File 1: `pareto-media-transform-matrix.csv`

The table below summarizes the dimensions of the MMM dataset used in this project.

Table 4.32: Dimensions of the MMM dataset

Attribute	Value
Number of rows	228872
Number of columns	12

The table below lists all column names along with their respective data types.

Table 4.33: Column names and data types in the dataset

Column Name	Data Type
...1	int64
ds	object
Digital_Weighted	object
TV_Like_Weighted	object
SEM_TimeWeighted	object
BRAND	object
INFLU_campaigns	object
GAMEPR_campaigns	object
type	object
solID	object
cluster	float64
top_sol	object

The table below shows the number of missing values per column.

Table 4.34: Missing values per column

Column Name	Missing Values
...1	0
ds	0
Digital_Weighted	0
TV_Like_Weighted	0
SEM_TimeWeighted	0
BRAND	32696
INFLU_campaigns	32696
GAMEPR_campaigns	32696
type	0
solID	0
cluster	3752
top_sol	3752

The following table shows descriptive statistics for the numerical columns.

Table 4.35: Descriptive statistics of numerical features

Feature	Mean	Std	Min	25%	50%
...1	114436.50	66069.80	1.00	57218.75	114436.50
cluster	5.18	2.96	1.00	2.75	5.00

The table below provides descriptive statistics for categorical columns.

Table 4.36: Summary statistics of categorical features

Feature	Count	Unique	Most Frequent	Frequency
ds	228872	268	2023-12-02	854
Digital_Weighted	228872	78172	11261,42119	610
TV_Like_Weighted	228872	45622	0	95648
SEM_TimeWeighted	228872	78439	11090,0472	610
BRAND	196176	75229	224,4897221	488
INFLU_campaigns	196176	21507	0	159513
GAMEPR_campaigns	196176	4962	0	179122
type	228872	7	rawMedia	32696
solID	228872	122	2_163_9	1876
top_sol	225120	2	False	206360

Below is a sample of the first five rows of the dataset.

Table 4.37: First five rows of the dataset

ds	Digital_Weighted	TV_Like_Weighted	SEM_TimeWeighted	BRAND	type	cluster
2023-03-10	11261,42119	0	11090,0472	224,4897221	rawMedia	3.0
2023-03-11	11878,61783	0	11986,08038	211,3344311	rawMedia	3.0
2023-03-12	11826,7793	0	12587,65975	276,9796393	rawMedia	3.0
2023-03-13	19801,39668	0	20915,19439	306,4889705	rawMedia	3.0
2023-03-14	13407,6863	0	21963,40369	231,6684644	rawMedia	3.0

4.12.7 Visual Overview of the Database

Tables (7)

Nom	Type	Schéma
Combined_CSVs.xlsx - pareto_aggregated		CREATE TABLE "Combined_CSVs.xlsx - pareto_aggregated" ("...1" INTEGER, "rn" TEXT, "coef" TEXT, "xDecompAgg" TEXT, "xDecompPerc" TEXT, "xDecompMeanNon0" TEXT, "xDecompMeanNon0Perc" TEXT, "xDecompAggRF" TEXT, "xDecompPercRF" TEXT, "xDecompMeanNon0RF" TEXT, "xDecompMeanNon0PercRF" TEXT, "pos" INTEGER, "train_size" TEXT, "rsq_train" TEXT, "rsq_val" TEXT, "rsq_test" TEXT, "nrmse_train" TEXT, "nrmse_val" TEXT, "nrmse_test" TEXT, "nrmse" TEXT, "decomp.rssd" TEXT, "mape" INTEGER, "lambda" TEXT, "lambda_hp" TEXT, "lambda_max" TEXT, "lambda_min_ratio" TEXT, "solid" TEXT, "trial" INTEGER, "iterNG" INTEGER, "iterPar" INTEGER, "iterations" INTEGER, "robynPareto" INTEGER, "total_spend" TEXT, "mean_spend" TEXT, "mean_spend_adstocked" TEXT, "mean_carryover" TEXT, "mean_response" TEXT, "spend_share" TEXT, "effect_share" TEXT, "roi_mean" TEXT, "roi_total" TEXT, "cpa_total" TEXT, "cluster" REAL, "top_sol" INTEGER, "boot_mean" TEXT, "boot_se" TEXT, "ci_low" TEXT, "ci_up" TEXT, "carryover_pot" TEXT)
...1	INTEGER	"...1" INTEGER
rn	TEXT	"rn" TEXT
coef	TEXT	"coef" TEXT
xDecompAgg	TEXT	"xDecompAgg" TEXT
xDecompPerc	TEXT	"xDecompPerc" TEXT
xDecompMeanNon0	TEXT	"xDecompMeanNon0" TEXT
xDecompMeanNon0Perc	TEXT	"xDecompMeanNon0Perc" TEXT
xDecompAggRF	TEXT	"xDecompAggRF" TEXT
xDecompPercRF	TEXT	"xDecompPercRF" TEXT
xDecompMeanNon0RF	TEXT	"xDecompMeanNon0RF" TEXT
xDecompMeanNon0PercRF	TEXT	"xDecompMeanNon0PercRF" TEXT
pos	INTEGER	"pos" INTEGER
train_size	TEXT	"train_size" TEXT

train_size	TEXT	"train_size" TEXT
rsq_train	TEXT	"rsq_train" TEXT
rsq_val	TEXT	"rsq_val" TEXT
rsq_test	TEXT	"rsq_test" TEXT
nrmse_train	TEXT	"nrmse_train" TEXT
nrmse_val	TEXT	"nrmse_val" TEXT
nrmse_test	TEXT	"nrmse_test" TEXT
nrmse	TEXT	"nrmse" TEXT
decomp.rssd	TEXT	"decomp.rssd" TEXT
mape	INTEGER	"mape" INTEGER
lambda	TEXT	"lambda" TEXT
lambda_hp	TEXT	"lambda_hp" TEXT
lambda_max	TEXT	"lambda_max" TEXT
lambda_min_ratio	TEXT	"lambda_min_ratio" TEXT
solID	TEXT	"solID" TEXT
trial	INTEGER	"trial" INTEGER
iterNG	INTEGER	"iterNG" INTEGER
iterPar	INTEGER	"iterPar" INTEGER
iterations	INTEGER	"iterations" INTEGER
robynPareto	INTEGER	"robynPareto" INTEGER
total_spend	TEXT	"total_spend" TEXT
mean_spend	TEXT	"mean_spend" TEXT
mean_spend_adstocked	TEXT	"mean_spend_adstocked" TEXT
mean_carryover	TEXT	"mean_carryover" TEXT

Nom	Type	Schéma
mean_response	TEXT	"mean_response" TEXT
spend_share	TEXT	"spend_share" TEXT
effect_share	TEXT	"effect_share" TEXT
roi_mean	TEXT	"roi_mean" TEXT
roi_total	TEXT	"roi_total" TEXT
cpa_total	TEXT	"cpa_total" TEXT
cluster	REAL	"cluster" REAL
top_sol	INTEGER	"top_sol" INTEGER
boot_mean	TEXT	"boot_mean" TEXT
boot_se	TEXT	"boot_se" TEXT
ci_low	TEXT	"ci_low" TEXT
ci_up	TEXT	"ci_up" TEXT
carryover_pct	TEXT	"carryover_pct" TEXT
Combined_CSVs.xlsx - pareto_alldecomp_matri x		CREATE TABLE "Combined_CSVs.xlsx - pareto_alldecomp_matrix" ("...1" INTEGER, "ds" TEXT, "dep_var" TEXT, "weekday" TEXT, "GT" TEXT, "CP_campaigns" TEXT, "obon230812" TEXT, "GW230429" TEXT, "GW230430" TEXT, "Digital_Weighted" TEXT, "TV_Like_Weighted" TEXT, "SEM_TimeWeighted" TEXT, "BRAND" TEXT, "INFLU_campaigns" TEXT, "GAMEPR_campaigns" TEXT, "intercept" TEXT, "depVarHat" TEXT, "solID" TEXT, "cluster" REAL, "top_sol" INTEGER)
...1	INTEGER	"...1" INTEGER
ds	TEXT	"ds" TEXT
dep_var	TEXT	"dep_var" TEXT
weekday	TEXT	"weekday" TEXT
GT	TEXT	"GT" TEXT
CP_campaigns	TEXT	"CP_campaigns" TEXT
obon230812	TEXT	"obon230812" TEXT
GW230429	TEXT	"GW230429" TEXT
GW230430	TEXT	"GW230430" TEXT
Digital_Weighted	TEXT	"Digital_Weighted" TEXT
TV_Like_Weighted	TEXT	"TV_Like_Weighted" TEXT

TV_Like_Weighted	TEXT	"TV_Like_Weighted" TEXT
SEM_TimeWeighted	TEXT	"SEM_TimeWeighted" TEXT
BRAND	TEXT	"BRAND" TEXT
INFLU_campaigns	TEXT	"INFLU_campaigns" TEXT
GAMEPR_campaigns	TEXT	"GAMEPR_campaigns" TEXT
intercept	TEXT	"intercept" TEXT
depVarHat	TEXT	"depVarHat" TEXT
solID	TEXT	"solID" TEXT
cluster	REAL	"cluster" REAL
top_sol	INTEGER	"top_sol" INTEGER
Combined_CSVs.xlsx - pareto_clusters		CREATE TABLE "Combined_CSVs.xlsx - pareto_clusters" ("...1" INTEGER, "solID" TEXT, "mape" INTEGER, "decomp.read" TEXT, "nrmse" TEXT, "nrmse test" TEXT, "nrmse train" TEXT, "nrmse_val" TEXT, "BRAND_thetas" TEXT, "Digital_Weighted_thetas" TEXT, "GAMEPR_campaigns_thetas" TEXT, "INFLU_campaigns_thetas" TEXT, "SEM_TimeWeighted_thetas" TEXT, "TV_Like_Weighted_thetas" TEXT, "BRAND_alphas" TEXT, "Digital_Weighted_alphas" TEXT, "GAMEPR_campaigns_alphas" TEXT, "INFLU_campaigns_alphas" TEXT, "SEM_TimeWeighted_alphas" TEXT, "TV_Like_Weighted_alphas" TEXT, "BRAND_gammas" TEXT, "Digital_Weighted_gammas" TEXT, "GAMEPR_campaigns_gammas" TEXT, "INFLU_campaigns_gammas" TEXT, "SEM_TimeWeighted_gammas" TEXT, "TV_Like_Weighted_gammas" TEXT, "cluster" INTEGER, "n" INTEGER, "top_sol" INTEGER)

Nom	Type	Schéma
...1	INTEGER	"...1" INTEGER
solid	TEXT	"solid" TEXT
mape	INTEGER	"mape" INTEGER
decomp.rssd	TEXT	"decomp.rssd" TEXT
nrmse	TEXT	"nrmse" TEXT
nrmse_test	TEXT	"nrmse_test" TEXT
nrmse_train	TEXT	"nrmse_train" TEXT
nrmse_val	TEXT	"nrmse_val" TEXT
BRAND_thetas	TEXT	"BRAND_thetas" TEXT
Digital_Weighted_thetas	TEXT	"Digital_Weighted_thetas" TEXT
GAMEPR_campaigns_thetas	TEXT	"GAMEPR_campaigns_thetas" TEXT
INFLU_campaigns_thetas	TEXT	"INFLU_campaigns_thetas" TEXT
SEM_TimeWeighted_thetas	TEXT	"SEM_TimeWeighted_thetas" TEXT
TV_Like_Weighted_thetas	TEXT	"TV_Like_Weighted_thetas" TEXT
BRAND_alphas	TEXT	"BRAND_alphas" TEXT
Digital_Weighted_alphas	TEXT	"Digital_Weighted_alphas" TEXT
GAMEPR_campaigns_alphas	TEXT	"GAMEPR_campaigns_alphas" TEXT
INFLU_campaigns_alphas	TEXT	"INFLU_campaigns_alphas" TEXT
SEM_TimeWeighted_alphas	TEXT	"SEM_TimeWeighted_alphas" TEXT
TV_Like_Weighted_alphas	TEXT	"TV_Like_Weighted_alphas" TEXT
BRAND_gammas	TEXT	"BRAND_gammas" TEXT
Digital_Weighted_gammas	TEXT	"Digital_Weighted_gammas" TEXT
GAMEPR_campaigns_gammas	TEXT	"GAMEPR_campaigns_gammas" TEXT
INFLU_campaigns_gammas	TEXT	"INFLU_campaigns_gammas" TEXT
SEM_TimeWeighted_gammas	TEXT	"SEM_TimeWeighted_gammas" TEXT
TV_Like_Weighted_gammas	TEXT	"TV_Like_Weighted_gammas" TEXT
cluster	INTEGER	"cluster" INTEGER
n	INTEGER	"n" INTEGER
top_sol	INTEGER	"top_sol" INTEGER

Combined_CSVs.xlsx - pareto_clusters_ci

...1	INTEGER	"...1" INTEGER
m	TEXT	"m" TEXT
cluster	INTEGER	"cluster" INTEGER
n	INTEGER	"n" INTEGER
boot_mean	TEXT	"boot_mean" TEXT
boot_se	TEXT	"boot_se" TEXT
boot_ci	TEXT	"boot_ci" TEXT
ci_low	TEXT	"ci_low" TEXT
ci_up	TEXT	"ci_up" TEXT
sd	TEXT	"sd" TEXT
dist100	TEXT	"dist100" TEXT

Combined_CSVs.xlsx - pareto_hyperparameters

...1	INTEGER	"...1" INTEGER
BRAND_alphas	TEXT	"BRAND_alphas" TEXT
BRAND_gammas	TEXT	"BRAND_gammas" TEXT
BRAND_thetas	TEXT	"BRAND_thetas" TEXT
Digital_Weighted_alphas	TEXT	"Digital_Weighted_alphas" TEXT
Digital_Weighted_gammas	TEXT	"Digital_Weighted_gammas" TEXT
Digital_Weighted_thetas	TEXT	"Digital_Weighted_thetas" TEXT
GAMEPR_campaigns_alphas	TEXT	"GAMEPR_campaigns_alphas" TEXT
GAMEPR_campaigns_gammas	TEXT	"GAMEPR_campaigns_gammas" TEXT
GAMEPR_campaigns_thetas	TEXT	"GAMEPR_campaigns_thetas" TEXT
INFLU_campaigns_alphas	TEXT	"INFLU_campaigns_alphas" TEXT
INFLU_campaigns_gammas	TEXT	"INFLU_campaigns_gammas" TEXT
INFLU_campaigns_thetas	TEXT	"INFLU_campaigns_thetas" TEXT
SEM_TimeWeighted_alphas	TEXT	"SEM_TimeWeighted_alphas" TEXT
SEM_TimeWeighted_gammas	TEXT	"SEM_TimeWeighted_gammas" TEXT

Nom	Type	Schéma
		TEXT, "SEM_TimeWeighted_thetas" TEXT, "TV_Like_Weighted_alphas" TEXT, "TV_Like_Weighted_gammas" TEXT, "TV_Like_Weighted_thetas" TEXT, "train_size" TEXT, "rsq_train" TEXT, "rsq_val" TEXT, "rsq_test" TEXT, "nrmse_train" TEXT, "nrmse_val" TEXT, "nrmse_test" TEXT, "nrmse" TEXT, "decomp.rssd" TEXT, "mape" INTEGER, "lambda" TEXT, "lambda_hp" TEXT, "lambda_max" TEXT, "lambda_min_ratio" TEXT, "pos" INTEGER, "Elapsed" TEXT, "ElapsedAccum" TEXT, "solID" TEXT, "trial" INTEGER, "iterNG" INTEGER, "iterPar" INTEGER, "iterations" INTEGER, "coef0" INTEGER, "mape.gt10" INTEGER, "robynPareto" INTEGER, "error_score" TEXT, "cluster" REAL, "top_sol" INTEGER)
...1	INTEGER	"...1" INTEGER
BRAND_alphas	TEXT	"BRAND_alphas" TEXT
BRAND_gammas	TEXT	"BRAND_gammas" TEXT
BRAND_thetas	TEXT	"BRAND_thetas" TEXT
Digital_Weighted_alphas	TEXT	"Digital_Weighted_alphas" TEXT
Digital_Weighted_gammas	TEXT	"Digital_Weighted_gammas" TEXT
Digital_Weighted_thetas	TEXT	"Digital_Weighted_thetas" TEXT
GAMEPR_campaigns_alphas	TEXT	"GAMEPR_campaigns_alphas" TEXT
GAMEPR_campaigns_gammas	TEXT	"GAMEPR_campaigns_gammas" TEXT
GAMEPR_campaigns_thetas	TEXT	"GAMEPR_campaigns_thetas" TEXT
INFLU_campaigns_alphas	TEXT	"INFLU_campaigns_alphas" TEXT
INFLU_campaigns_gammas	TEXT	"INFLU_campaigns_gammas" TEXT
INFLU_campaigns_thetas	TEXT	"INFLU_campaigns_thetas" TEXT
SEM_TimeWeighted_alphas	TEXT	"SEM_TimeWeighted_alphas" TEXT
SEM_TimeWeighted_gammas	TEXT	"SEM_TimeWeighted_gammas" TEXT
SEM_TimeWeighted_thetas	TEXT	"SEM_TimeWeighted_thetas" TEXT
TV_Like_Weighted_alphas	TEXT	"TV_Like_Weighted_alphas" TEXT
TV_Like_Weighted_gammas	TEXT	"TV_Like_Weighted_gammas" TEXT
TV_Like_Weighted_thetas	TEXT	"TV_Like_Weighted_thetas" TEXT
train_size	TEXT	"train_size" TEXT

train_size	TEXT	"train_size" TEXT
rsq_train	TEXT	"rsq_train" TEXT
rsq_val	TEXT	"rsq_val" TEXT
rsq_test	TEXT	"rsq_test" TEXT
nrmse_train	TEXT	"nrmse_train" TEXT
nrmse_val	TEXT	"nrmse_val" TEXT
nrmse_test	TEXT	"nrmse_test" TEXT
nrmse	TEXT	"nrmse" TEXT
decomp.rssd	TEXT	"decomp.rssd" TEXT
mape	INTEGER	"mape" INTEGER
lambda	TEXT	"lambda" TEXT
lambda_hp	TEXT	"lambda_hp" TEXT
lambda_max	TEXT	"lambda_max" TEXT
lambda_min_ratio	TEXT	"lambda_min_ratio" TEXT
pos	INTEGER	"pos" INTEGER
Elapsed	TEXT	"Elapsed" TEXT
ElapsedAccum	TEXT	"ElapsedAccum" TEXT
solID	TEXT	"solID" TEXT
trial	INTEGER	"trial" INTEGER
iterNG	INTEGER	"iterNG" INTEGER
iterPar	INTEGER	"iterPar" INTEGER
iterations	INTEGER	"iterations" INTEGER
coef0	INTEGER	"coef0" INTEGER

Nom	Type	Schéma
mape.qt10	INTEGER	"mape.qt10" INTEGER
robynPareto	INTEGER	"robynPareto" INTEGER
error_score	TEXT	"error_score" TEXT
cluster	REAL	"cluster" REAL
top_sol	INTEGER	"top_sol" INTEGER
Combined_CSvs.xlsx - pareto_media_transform _matrix		CREATE TABLE "Combined_CSvs.xlsx - pareto_media_transform_matrix" ("...1" INTEGER, "ds" TEXT, "Digital_Weighted" TEXT, "TV_Like_Weighted" TEXT, "SEM_TimeWeighted" TEXT, "BRAND" TEXT, "INFLU_campaigns" TEXT, "GAMEPR_campaigns" TEXT, "type" TEXT, "solid" TEXT, "cluster" REAL, "top_sol" INTEGER)
...1	INTEGER	"...1" INTEGER
ds	TEXT	"ds" TEXT
Digital_Weighted	TEXT	"Digital_Weighted" TEXT
TV_Like_Weighted	TEXT	"TV_Like_Weighted" TEXT
SEM_TimeWeighted	TEXT	"SEM_TimeWeighted" TEXT
BRAND	TEXT	"BRAND" TEXT
INFLU_campaigns	TEXT	"INFLU_campaigns" TEXT
GAMEPR_campaigns	TEXT	"GAMEPR_campaigns" TEXT
type	TEXT	"type" TEXT
solid	TEXT	"solid" TEXT
cluster	REAL	"cluster" REAL
top_sol	INTEGER	"top_sol" INTEGER

Combined_CSvs.xlsx - raw_data		CREATE TABLE "Combined_CSvs.xlsx - raw_data" ("...1" INTEGER, "DATE" TEXT, "CV" TEXT, "SEM" TEXT, "ADNW1" TEXT, "ADNW2" INTEGER, "ADNW3" TEXT, "SNS" TEXT, "AFF" TEXT, "DOUGA" TEXT, "BRAND" TEXT, "GT" INTEGER, "SEM_lag1" TEXT, "SEM_lag2" TEXT, "Digital_Weighted" TEXT, "TV_Like_Weighted" TEXT, "SEM_TimeWeighted" TEXT, "GAMEPR230324" INTEGER, "GAMEPR230727" INTEGER, "INFLU230402" INTEGER, "INFLU230404" INTEGER, "INFLU230408" INTEGER, "INFLU230821" INTEGER, "CP230317" INTEGER, "CP230421" INTEGER, "CP230811" INTEGER, "obon230812" INTEGER, "GW230429" INTEGER, "GW230430" INTEGER, "INFLU_campaigns" INTEGER, "GAMEPR_campaigns" INTEGER, "CP_campaigns" INTEGER)
...1	INTEGER	"...1" INTEGER
DATE	TEXT	"DATE" TEXT
CV	TEXT	"CV" TEXT
SEM	TEXT	"SEM" TEXT
ADNW1	TEXT	"ADNW1" TEXT
ADNW2	INTEGER	"ADNW2" INTEGER
ADNW3	TEXT	"ADNW3" TEXT
SNS	TEXT	"SNS" TEXT
AFF	TEXT	"AFF" TEXT
DOUGA	TEXT	"DOUGA" TEXT
BRAND	TEXT	"BRAND" TEXT
GT	INTEGER	"GT" INTEGER
SEM_lag1	TEXT	"SEM_lag1" TEXT
SEM_lag2	TEXT	"SEM_lag2" TEXT
Digital_Weighted	TEXT	"Digital_Weighted" TEXT

Digital_Weighted	TEXT	"Digital_Weighted" TEXT
TV_Like_Weighted	TEXT	"TV_Like_Weighted" TEXT
SEM_TimeWeighted	TEXT	"SEM_TimeWeighted" TEXT
GAMEPR230324	INTEGER	"GAMEPR230324" INTEGER
GAMEPR230727	INTEGER	"GAMEPR230727" INTEGER
INFLU230402	INTEGER	"INFLU230402" INTEGER

Nom	Type	Schéma
INFLU230404	INTEGER	"INFLU230404" INTEGER
INFLU230408	INTEGER	"INFLU230408" INTEGER
INFLU230821	INTEGER	"INFLU230821" INTEGER
CP230317	INTEGER	"CP230317" INTEGER
CP230421	INTEGER	"CP230421" INTEGER
CP230811	INTEGER	"CP230811" INTEGER
obon230812	INTEGER	"obon230812" INTEGER
GW230429	INTEGER	"GW230429" INTEGER
GW230430	INTEGER	"GW230430" INTEGER
INFLU_campaigns	INTEGER	"INFLU_campaigns" INTEGER
GAMEPR_campaigns	INTEGER	"GAMEPR_campaigns" INTEGER
CP_campaigns	INTEGER	"CP_campaigns" INTEGER

Index (0)

Nom	Type	Schéma
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Vues (0)

Nom	Type	Schéma
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Déclencheurs (0)

Nom	Type	Schéma
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